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SYSTEMATIC REVIEW

The Association Between Training Load and Performance in Team Sports: A Systematic Review

Jordan L. Fox[1,2] · Robert Stanton[1,3] · Charli Sargent[3] · Sally-Anne Wintour[1] · Aaron T. Scanlan[1,2]

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Abstract

Background Adequate training loads promote favorable physical and physiological adaptations, reduce the likelihood of illness and injury, and, therefore, increase the possibility of success during competition.

Objectives Our objective was to systematically examine the association between training load and performance outcomes in team sports.

Methods We systematically searched the PubMed, SPORTDiscus, and PsycINFO databases for original research published before July 2018. The search included terms relevant to training load, performance, and team sports. Articles were screened using pre-defined selection criteria, and methodological quality was assessed independently by two authors before data were extracted by the lead author.

Results The electronic search yielded 5848 articles, 2373 of which were duplicates. A further 17 articles were retrieved from additional sources. In total, 26 articles met the inclusion criteria for this review, with quality scores ranging from 6 to 10 out of 11. Training exposure was more strongly associated with aerobic performance than other external training load measures. High-intensity activity (≥ 90% of maximum heart rate) was strongly associated with aerobic performance. The individualized training impulse model was strongly associated with aerobic performance, whereas various other training impulse models and perceptual training load measures showed weak associations with aerobic performance. There were no clear associations between training load and neuromuscular variables or game-related statistics.

Conclusion We found no consistent associations between external training load measures and performance. High-intensity internal training load appears to be the most prominent indicator of aerobic performance.

Key Points

There is strong evidence to support the associations between heart rate-mediated high-intensity internal training load and changes in aerobic variables in field-based team sports.

Team sport coaches and practitioners should exercise caution when using external training load measures, total and differential session rating of perceived exertion, and Banister's, Edwards', and Lucia's training impulse models to induce anticipated changes in performance in players.

There is a lack of consistent evidence supporting an association between external training load and performance as indicated by changes in aerobic or neuromuscular variables and game-related statistics in team sports.

✉ Jordan L. Fox
j.fox2@cqu.edu.au

1 School of Health, Medical and Applied Sciences, Central Queensland University, Rockhampton, QLD 4702, Australia

2 Human Exercise and Training Laboratory, Central Queensland University, Rockhampton, QLD, Australia

3 Appleton Institute for Behavioural Science, Central Queensland University, Wayville, SA, Australia

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1 Introduction

Training strategies in team sports are implemented with the aim of promoting favorable performance outcomes for players and teams during competition while reducing the likelihood of illness, injury, and maladaptive responses to training [1]. To achieve these goals, training is often periodized across seasonal phases of the annual plan to encourage peak performance to coincide with competition schedules [2]. For a periodized training approach to be effective, it should involve the prescription of appropriate training loads dispersed between sufficient recovery interventions [1, 3]. Effectively prescribing and monitoring training loads for players requires a balance between external and internal training loads. The relationship between external and internal training loads represents the dose–response nature of the training stimulus imposed and the resultant player responses [4–6]. External training load represents the physical workload encountered by the player (i.e., the stimulus imposed) and is often measured as exposure time (either total duration or time spent completing individual activities), speed, distance, movement volume, or movement intensity obtained via microsensors and video-based time–motion analysis [7–9]. Internal training load represents the biomechanical, physical, and physiological responses of players to training or competition stimuli [10]. Internal training load is typically assessed using perceptual measures, heart rate (HR)-derived training load models, or time spent working in HR-mediated intensity zones [3, 9, 11].

Training load monitoring in team sports can be implemented for a number of reasons, such as monitoring fatigue, identifying injury risk, and determining player readiness to perform [1, 10]. Training load measures are used in a multitude of ways, but there appears to be no clear consensus within current literature on the most appropriate method for monitoring training loads in team sport. Given training load measures can be used to anticipate a range of outcomes, it is necessary to consider the relationships between training load measures and specific variables (e.g., fatigue, injury risk, or performance). Several recent literature reviews [12–14] have examined the association between training load and injury in team sports. A recent review by Jaspers et al. [15] examined the association between training load indicators and training outcomes in professional soccer players. However, no literature has synthesized original research describing the association between training load and performance in team sports. Addressing performance outcomes specifically, rather than training outcomes, may be of greater interest to coaches and practitioners and advance the application of research to practice. Identifying the associations between training load and various performance outcomes in team sports is important in developing a fundamental understanding of what training load approaches and thresholds can be most effectively implemented to induce optimal performance gains. This knowledge may permit coaching staff and practitioners to adopt relevant training load models and modify training loads to elicit the desired performance outcomes while limiting the likelihood of poor training adaptations that occur as a result of insufficient or excessive training stimuli [1, 10, 16]. As such, our aim was to conduct a systematic review of published literature to identify the associations between training load and performance measures in team sports.

2 Methods

2.1 Literature Search Strategy

Articles were systematically identified via three electronic databases (PubMed, SPORTDiscus, and PsycINFO) using the search strategy presented in Table 1. The search string for each variable (training load, performance, and team sports) was used independently, after which all three were combined in the complete search strategy. The search was restricted to original peer-reviewed studies published in English, with literature reviews and conference proceedings excluded. The search was developed to consider research articles published online or in print from database inception until July 2018, when the search was conducted.

Table 1 Search strategy used to locate relevant research articles

Variable	Search terms
Training load variable	('training load' OR 'training impulse' OR TRIMP OR 'external load' OR 'internal load' OR 'duration' OR 'exposure' OR 'RPE' OR 'rating of perceived exertion' OR 'summated-heart-rate-zone' OR 'SHRZ' OR 'PlayerLoad' OR 'BodyLoad' OR 'global positioning system' OR 'GPS' or 'accelerometer')
Performance variable	(performance OR fitness OR adaptation OR game OR match OR competition OR capacity* OR ability*)
Team sports	('team sport' OR 'court sport' OR 'field sport' OR 'team')
Training load AND performance AND team sports (final search)	'1 AND 2 AND 3'

2.2 Selection Criteria

The process for screening articles followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [17]. Articles considered for inclusion in the review were those examining team sports, reporting training load and performance outcomes, and using statistical analyses to quantify the association between training load and performance measures. No restrictions were imposed on the participant sample (e.g., age, sex, playing level, or playing experience). Including experimental studies that implemented an intervention may have misrepresented any associations between training load and performance, so the review was restricted to cross-sectional or longitudinal observational study designs.

For the purpose of this review, training load was categorized as external training load or internal training load. External training load was defined as the volume, intensity, or frequency of the training or competition stimuli imposed on players (e.g., the amount of movement completed). Internal training load was defined as player responses to the imposed stimulus (e.g., physical, physiological, or perceptual stress). Performance was considered to be any objective indicator of in-game performance (e.g., game-related statistics) or changes in physical or physiological assessments across the monitoring period (e.g., aerobic fitness). Studies where player monitoring data were reported only during competitive games were excluded as they did not represent the complete workloads experienced by players.

The abstracts of all of the articles identified in the search were screened independently against the pre-defined selection criteria by two authors (JLF and SAW). Any disagreements between the two authors regarding article inclusion were further discussed and, if agreement was not reached, a third author (ATS) was consulted to establish consensus. Full-text versions of the included articles were then obtained and screened using the same process. The reference list of each included article was hand searched for any suitable articles not identified in the original search. Despite the systematic search strategy, a relevant article was not identified in the original search or via the reference lists of included articles, possibly due to discrepancies in the terminology used to describe training load measures. To ensure this review was as comprehensive as possible, this article was subsequently included and is grouped in Fig. 1 with "records retrieved via additional sources." This approach has been previously adopted in reviews examining training load studies [13].

2.3 Assessment of Methodological Quality

Methodological quality was assessed using a modified version of the Downs and Black [18] checklist for assessing the methodological quality of randomized and nonrandomized healthcare interventions. We chose this checklist because it has been validated for use with observational study designs [18] and has been previously used to assess methodological quality in systematic reviews assessing cross-sectional and longitudinal studies [19–21]. The number of items from the original checklist can be tailored to the scope and needs of the systematic review, with 10–15 items used in previous systematic reviews [19–21]. For this review, 11 items in the checklist were deemed relevant (see Table 2). Each item is scored as "1" (yes) or

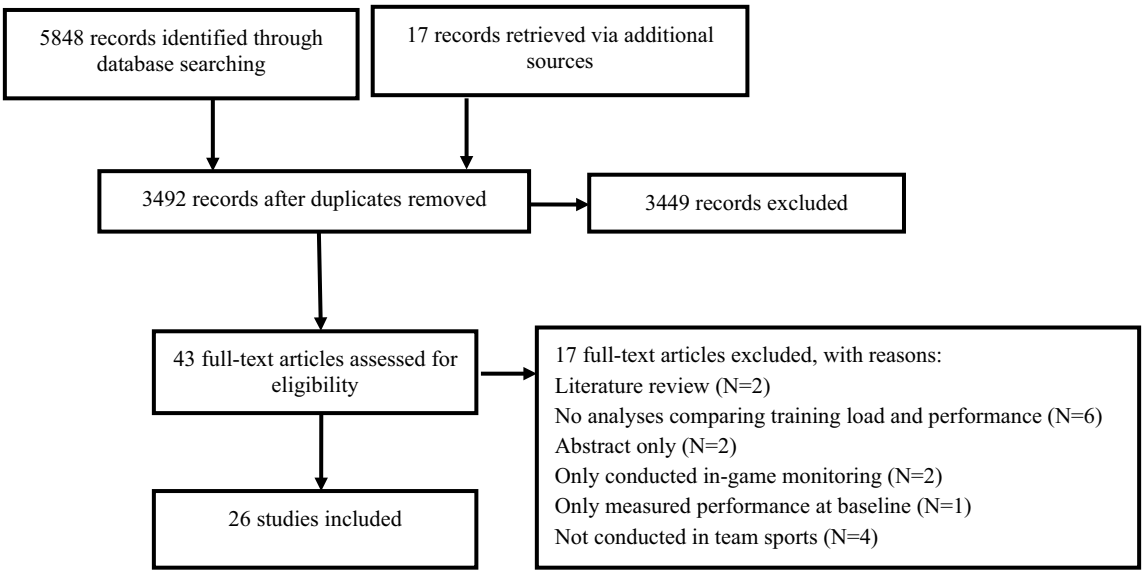

Fig. 1 Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) flow diagram of search strategy

Table 2 Questions from the modified Downs and Black checklist [18] used to evaluate methodological quality of the included articles

Question no.	Question
	Reporting
1	Is the hypothesis/aim/objective of the study clearly described?
2	Are the main outcomes to be measured clearly described in the introduction or methods section?
3	Are the characteristics of the subjects included in the study clearly described?
4	Are the main findings of the study clearly described?
5	Does the study provide estimates of the random variability in the data for the main outcomes?
6	Have actual probability values been reported (e.g., 0.035 rather than < 0.05) for the main outcomes except where the probability value is < 0.001?
	External validity
7	Were the subjects asked to participate in the study representative of the entire population from which they were recruited?
8	Were those subjects who were prepared to participate representative of the entire population from which they were recruited?
	Internal validity bias
9	If any of the results of the study were based on "data dredging," was this made clear?
10	Were the statistical tests used to assess the main outcomes appropriate?
11	Were the main outcome measures accurate (valid and reliable)?

"0" (no/unable to determine), and the scores for each of the 11 items are summed to provide the total quality score. The quality of each included article was rated against the checklist independently by two authors (JLF and SAW). Any disparity in the outcome of the quality appraisal was discussed, and a third author (ATS) was consulted if a decision could not be reached.

2.4 Data Extraction and Analysis

Data were extracted from each article by the lead author (JLF). Data not provided or presented non-numerically were identified as "not reported." Where items pertaining to study results were not reported (e.g., studies where association statistics and probability values were reported for significant results only), these data were omitted to make the tables more concise. The following data, where possible, were extracted from each article:

- Participants' characteristics, sport investigated, sample size, playing level, sex, age, stature, and body mass
- Study methodology – the monitoring period (i.e., seasonal phase(s) and duration), training load measures, and performance measures (i.e., outcome measures, tests administered, and timing of assessments)
- Study results – training load and performance measures compared and the results of statistical analyses (i.e., association statistic, interpretation, and statistical significance). Given the variability in cutpoints used to describe the magnitude of the association statistics, interpretations were only provided if reported in the articles. This approach avoided bias in the reporting of results.

The outcome measures reported and the statistical tests adopted between studies was inconsistent, so data could not be pooled, and a meta-analysis was not performed. Where possible, participants' characteristics are reported as mean ± standard deviation. Given the range of outcome measures reported in each of the studies, formulae for relevant training load and performance measures are presented in Table 3.

3 Results

3.1 Search Findings and Study Selection

The electronic search yielded 5848 articles (PubMed = 2986, SPORTDiscus = 2127, PsycINFO = 735). An additional 17 articles from additional sources were identified as potentially relevant. A total of 2373 duplicate records were removed, and a further 3449 irrelevant articles were excluded based on title and abstract; 43 full-text articles were screened and 17 were removed, leaving 26 articles for inclusion in the review. Reasons for exclusion were article type (literature review $N = 2$; conference proceeding $N = 2$), failure to perform statistical analyses between training load measures and performance measures ($N = 6$), player monitoring limited to competitive games only ($N = 2$), performance assessments conducted at baseline only ($N = 1$), and examination of individual sports rather than team sports ($N = 4$). The full results of the search are presented in Fig. 1.

Table 3 Formulae used to calculate training load outcome measures in the included articles

Outcome measure	Formula	Abbreviations and definitions
Ranking	$(SP+R+A+ST+BM)-(MS+T+BA+FC)$	SP = scored points, R = total rebounds (offensive and defensive), A = assists, ST = steals, BM = blocks made, MS = missed shots (2 points, 3 points, and free throws), T = turnovers, BA = blocks attempted, and FC = faults incurred
PlayerLoad™	$[\sqrt{(Ac1_n-Ac1_{n-1})^2+(Ac2_n-Ac2_{n-1})^2+(Ac3_n-Ac3_{n-1})^2}]\times 0.01$	Ac1, Ac2 and Ac3 are the orthogonal components measured from the tri-axial accelerometer and 0.01 is the scaling factor
Champion Data© player rank	A proprietary, undisclosed algorithm is used to calculate the Champion Data© player rank	Champion Data© player rank is based on official Australian Football League statistics that players accumulate during a game and is weighted towards efficient ball use and gaining possession in congested situations [30]
sRPE	$sRPE=RPE\times duration$	RPE = Borg's 1–10 category ratio scale and duration = time in minutes
Banister's TRIMP	$Load=[duration\ (min)](HR_{ex}-HR_{rest})/(HR_{max}-HR_{rest})\times 0.64e^{1.92x}$	HR_{ex} = average HR during exercise, HR_{rest} = HR at rest, HR_{max} = maximal HR, e = 2.712, and $_x=(HR_{ex}-HR_{rest})/(HR_{max}-HR_{rest})$
Team TRIMP	$Load=duration\times\Delta HR\times 0.2053e^{3.5179x}$	Duration = time in minutes, $\Delta HR=HR_{exercise}-HR_{rest}/HR_{max}-HR_{rest}$, e = the base of the natural logarithms, x = ΔHR, and 0.2053 and 3.5179 are constants
iTRIMP	$Load=Duration\times\Delta HR\times yi$	$\Delta HR=HR_{exercise}$ - $HR_{rest}/HR_{max}-HR_{rest}$, yi = individualized weighting factor determined during the maximal exercise test
Training–stress balance	$Balance=weekly\ load/monthly\ load$	Weekly load = mean weekly sRPE and monthly load = mean monthly sRPE
Training strain	$Strain=mean\ weekly\ training\ load\times monotony$	Monotony = mean weekly training load/standard deviation
Lucia's TRIMP	$Load=(duration\ in\ zone\ 1\times 1)+(duration\ in\ zone\ 2\times 2)+(duration\ in\ zone\ 3\times 3)$	Zone 1 = HR corresponding with $[BLa^{-1}]<2.5$ mmol·L^{-1}, zone 2 = HR corresponding with $[BLa^{-1}]\geq 2.5$ mmol·L^{-1} and ≤ 4.0 mmol·L^{-1} and zone 3 = HR corresponding with $[BLa^{-1}]>4.0$ mmol•L^{-1}
Edwards' TRIMP	$Load=(duration\ in\ zone\ 1\times 1)+(duration\ in\ zone\ 2\times 2)+(duration\ in\ zone\ 3\times 3)+(duration\ in\ zone\ 4\times 4)+(duration\ in\ zone\ 5\times 5)$	Zone 1 = 50–59.9% HR_{max}, zone 2 = 60–69.9% HR_{max}, zone 3 = 70–79.9% HR_{max}, zone 4 = 80–89.9% HR_{max}, and zone 5 = 90–100% HR_{max}

[BLa^{-1}] blood lactate concentration, *HR* heart rate, *iTRIMP* individualized training impulse, *sRPE* session rating of perceived exertion, *TRIMP* training impulse

3.2 Methodological Quality

The ratings from the quality appraisal for each article are presented in Table 4. Methodological quality scores ranged from 6 to 10 out of 11. In line with previous literature using the Downs and Black checklist [19–21], no articles were excluded on the basis of methodological quality.

3.3 Participant Characteristics

The characteristics of the participants investigated in the included articles are presented in Table 5. Participants monitored in each study competed in various sports, including Australian Rules football ($N=3$), basketball ($N=1$), field hockey ($N=1$), futsal ($N=1$), handball ($N=2$), hurling ($N=3$), rugby league ($N=2$), rugby union ($N=1$), volleyball ($N=1$), and soccer ($N=11$). Sample sizes ranged from 8 to 183 players and covered a range of playing levels. In total, 20 studies monitored only males, one monitored only females, and five did not report the sex of the participants.

3.4 Outcome Measures

In the included articles, training load was reported as external training load only ($N=3$), internal training load only ($N=14$),

Table 4 Results of methodological quality assessment for included articles

Study	Downs and Black [18] checklist question number											Total
	Reporting						External validity		Internal validity bias			
	1	2	3	4	5	6	7	8	9	10	11	
Akubat et al. [3]	1	1	1	1	1	1	0	1	1	1	1	**10**
Aughey et al. [42]	1	1	0	1	1	0	0	0	1	1	1	**7**
Brink et al. [22]	1	0	0	1	1	0	0	0	1	1	1	**6**
Campos-Vazquez et al. [7]	1	1	0	1	1	0	0	0	1	1	1	**7**
Caparros et al. [8]	1	1	1	1	1	1	0	0	1	1	1	**9**
Castagna et al. [31]	1	1	0	1	1	1	0	0	1	1	1	**8**
Castagna et al. [32]	1	1	1	1	1	1	0	0	1	1	1	**9**
Dobbin et al. [39]	1	1	1	1	1	0	0	0	1	1	1	**8**
Fitzpatrick et al. [27]	1	1	1	1	1	0	0	0	1	1	1	**8**
Gabbett and Domrow [37]	1	1	0	1	1	1	1	0	1	1	1	**9**
Gil-Rey et al. [23]	1	1	1	1	1	1	0	1	1	1	1	**10**
Gorostiaga et al. [28]	1	1	1	1	1	0	0	0	1	1	1	**8**
Granados et al. [29]	1	1	1	1	1	0	0	0	1	1	1	**8**
Harrison and Johnston [40]	1	1	1	1	1	1	0	0	1	1	1	**8**
Horta et al. [41]	1	1	1	1	1	1	0	1	1	1	1	**10**
Los Arcos et al. [24]	1	1	1	1	1	0	0	1	1	1	1	**9**
Los Arcos et al. [25]	1	1	1	1	1	1	0	1	1	1	1	**10**
Los Arcos et al. [26]	1	1	1	1	1	0	0	1	1	1	1	**9**
McCaskie et al. [30]	1	1	1	1	1	1	0	0	1	1	1	**9**
Malone and Collins [34]	1	1	1	1	1	1	0	0	1	1	1	**9**
Malone et al. [33]	1	1	0	1	1	1	0	0	1	1	1	**8**
Malone et al. [36]	1	1	1	1	1	0	0	0	1	1	1	**8**
Manzi et al. [35]	1	1	1	1	1	1	0	0	1	1	1	**9**
Nakamura et al. [38]	1	1	0	1	1	0	0	0	1	1	1	**7**
Stagno et al. [11]	1	1	1	1	1	1	0	0	1	1	1	**9**
Taylor et al. [9]	1	1	0	1	1	1	0	0	1	1	1	**8**

1 = yes; 0 = no/unable to determine

or external and internal training load ($N=9$). Training load data were collected during the preseason phase only ($N=9$), in-season phase only ($N=11$), or pre- and in-season phases ($N=6$). Performance outcomes were reported as either game-related statistics ($N=3$) or changes in physical or physiological assessments administered at various time points across the annual training plan not as a result of any intervention from the research team ($N=23$). In the 23 studies examining changes in physical or physiological assessments, performance was reported as aerobic variables only ($N=12$), neuromuscular variables only ($N=2$), or aerobic and neuromuscular variables ($N=9$). The methodological approach and outcome measures reported in each article are described in Table 6.

3.5 Associations Between External Training Load and Performance

3.5.1 External Training Load and Aerobic Performance

The associations between external training load and performance are presented in Table 7. One study [22] used prediction analyses to identify the association between external training load and performance. Brink et al. [22] collected total exposure (training and games) over 7 months and conducted an intermittent shuttle run test each month to measure endurance capacity. A multilevel regression analysis was conducted to determine whether exposure 1 and 2 weeks

Table 5 Characteristics of the participants examined in each included article

Study	Sport	Sample size (N)	Playing level	Sex	Age (years)	Stature (m)	Body mass (kg)
Akubat et al. [3]	Soccer	9	Professional, junior	Male	17.0 ± 1.0	1.81 ± 0.05	72.9 ± 6.7
Aughey et al. [42]	AFL	59	Elite	Male	24.7 ± 3.3	1.87 ± 0.77	90.4 ± 8.2
Brink et al. [22]	Soccer	18	Elite, junior	Male	17.0 ± 0.5	1.77 ± 0.62	72.4 ± 7.8
Campos-Vazquez et al. [7]	Soccer	12	Professional	Male	27.7 ± 4.3	1.77 ± 0.62	73.1 ± 5.2
Caparros et al. [8]	Basketball	44	Professional	Male	27.6 ± 0.8	2.00 ± 0.08	98.5 ± 2.3
Castagna et al. [31]	Soccer	14	Professional	Male	25.0 ± 4.0	1.78 ± 0.70	74.0 ± 8.0
Castagna et al. [32]	Soccer	18	Professional	Male	28.6 ± 3.2	1.83 ± 0.61	80.0 ± 5.4
Dobbin et al. [39]	Rugby League	16	Professional, junior	NR	17.2 ± 0.7	1.80 ± 0.49	88.5 ± 10.1
Fitzpatrick et al. [27]	Soccer	14	Professional, junior	Male	17.1 ± 0.5	1.78 ± 0.46	70.9 ± 5.8
Gabbett and Domrow [37]	Rugby league	183	Sub-elite	Male	21.4 ± 5.1	1.75 ± 0.07	85.8 ± 14.6
Gil-Rey et al. [23]	Soccer	28	Elite (N = 14) Sub-elite (N = 14)	Male	Elite: 17.6 ± 0.6 Sub-elite: 17.5 ± 0.5	Elite: 1.80 ± 0.56 Sub-elite: 1.78 ± 5.60	Elite: 70.3 ± 4.4 Sub-elite: 71.1 ± 6.5
Gorostiaga et al. [28]	Handball	15	Elite	Male	31.0 ± 3.0	1.88 ± 0.70	95.6 ± 14.3
Granados et al. [29]	Handball	16	Elite	Female	23.1 ± 4.0	NR	NR
Harrison and Johnston [40]	AFL	60	Semi-professional	NR	21.3 ± 2.9	1.85 ± 0.78	83.3 ± 9.0
Horta et al. [41]	Volleyball	12	Elite	Male	26.9 ± 4.6	1.95 ± 0.08	94.9 ± 11.6
Los Arcos et al. [24]	Soccer	21	Professional	Male	21.0 ± 1.7	1.81 ± 0.63	76.1 ± 7.7
Los Arcos et al. [25]	Soccer	14	Professional, junior	Male	20.6 ± 1.7	1.79 ± 0.06	73.5 ± 7.0
Los Arcos et al. [26]	Soccer	14	Professional	Male	20.6 ± 1.5	1.80 ± 0.05	73.6 ± 7.4
Malone and Collins [34]	Hurling	20	Elite	Male	25.5 ± 3.0	1.79 ± 3.20	78.5 ± 4.5
Malone et al. [33]	Hurling	30	Elite	NR	26.5 ± 3.2	1.79 ± 3.20	81.5 ± 4.5
Malone et al. [36]	Hurling	30	Elite	NR	24.0 ± 4.0	1.80 ± 0.02	78.0 ± 3.0
Manzi et al. [35]	Soccer	18	Professional	Male	28.4 ± 3.2	1.82 ± 5.30	79.9 ± 5.5
McCaskie et al. [30]	AFL	41	Professional	Male	23.0 ± 3.0	1.88 ± 0.07	86.7 ± 7.9
Nakamura et al. [38]	Futsal	10	Professional	NR	19.1 ± 0.8	1.75 ± 0.62	71.3 ± 6.6
Stagno et al. [11]	Field hockey	8	Elite	Male	24.0 ± 4.0	1.81 ± 0.04	80.8 ± 5.2
Taylor et al. [9]	Rugby union	10	Collegiate	Male	18.4 ± 1.0	1.81 ± 0.59	85.9 ± 13.0

AFL Australian Rules Football, *NR* not reported

Table 6 Methodological approaches of each of the included articles

Study	Monitoring period	Training load measures	Performance measures
Akubat et al. [3]	Training sessions and games for 6 weeks during the in-season	Mean weekly training load using Banister's TRIMP[a], Team TRIMP[a], iTRIMP[a], and sRPE[a] (AU)	Incremental treadmill test administered pre and post monitoring period to compare velocity at LT ($km \cdot h^{-1}$), HR at LT ($beats \cdot min^{-1}$), velocity at OBLA ($km \cdot h^{-1}$), and HR at OBLA ($beats \cdot min^{-1}$)
Aughey et al. [41]	Training sessions and games across six seasons	Mean weekly and monthly training load and training strain[a], and training–stress balance[a] (load and strain) using sRPE[a] (AU)	Game outcome (win/loss) with and without the covariates of ladder position and days between games
Brink et al. [22]	Training sessions and games for 7 months during the in-season	Mean weekly exposure time 1 and 2 weeks before each test (min) and mean weekly training load 1 week and 2 weeks before each test using sRPE[a] (AU)	Submaximal ISRT administered monthly to determine HR corresponding to 70% of maximal speed reached during test 1 ($beats \cdot min^{-1}$)
Campos-Vazquez et al. [7]	Training sessions and games for 4 weeks during the preseason	Training exposure (min), practice game exposure (min), mean training session duration (min), mean practice game duration (min), total training load using Edwards' TRIMP[a] (AU), and volume of high-intensity activity (> 90% of HR_{max}) (min)	30–15 IFT administered pre and post monitoring period to determine peak velocity ($km \cdot h^{-1}$)
Caparros et al. [8]	Training sessions and games during the in-season of seven consecutive seasons	Number of practices and games per season, practice hours, game hours, and total exposure hours per season (mean ± SD)	Total team ranking[a], mean game ranking[a], mean player ranking[a], and mean player points scored per season
Castagna et al. [31]	Training sessions for 6 weeks during the preseason	Time (min) spent completing low-, medium-, and high-intensity activity based on HR zones calculated using [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$	Maximal, incremental exercise test administered pre and post monitoring period to determine speeds at [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$ ($km \cdot h^{-1}$)
Castagna et al. [32]	Training sessions for 8 weeks during the preseason	Time (min) spent completing low-, medium-, and high-intensity activity based on HR zones calculated using [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$	Yo-Yo IRT (level 1) administered pre and post monitoring period to determine speed at [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$ ($km \cdot h^{-1}$) and VO_{2max} ($ml \cdot kg^{-1} \cdot min^{-1}$)
Dobbin et al. [39]	Training sessions for 14 weeks during the preseason	Resistance training load, conditioning training load, skills training load, and total training load using sRPE[a] (AU)	Battery of tests administered pre and post monitoring period to determine 10- and 20-m sprint times (s), 10- and 20-m sprint momentum ($kg \cdot m \cdot s^{-1}$), CMJ height (cm), COD test time (s), medicine ball throwing distance (m), and Yo-Yo IRT (level 1) performance (m)
Fitzpatrick et al. [28]	Training sessions and games for 6 weeks during the in-season	Mean weekly training load using total distance (m), AD load (acceleration and deceleration distance > 2 $m \cdot s^{-2}$) (m), m > HSD (distance covered above 17 $km \cdot h^{-1}$) (m), t > HSD (time spent above 17 $km \cdot h^{-1}$) (min), m > VHSD (distance covered above 21 $km \cdot h^{-1}$) (m), t > VHSD (time spent above 21 $km \cdot h^{-1}$) (min), m > MAS (m), t > MAS (min), m > 30ASR (m), t > 30ASR (min), Edwards TRIMP (AU), and sRPE (AU)	Battery of tests administered pre and post monitoring period to determine maximal sprint speed ($km \cdot h^{-1}$) and maximal aerobic speed ($km \cdot h^{-1}$)

Table 6 (continued)

Study	Monitoring period	Training load measures	Performance measures
Gabbett and Domrow [37]	Training sessions and games across two seasons split into preseason, early-competition, and late-competition phases	Mean weekly sRPE[a] (AU) during each of the seasonal phases investigated	Battery of tests administered during the off-season, preseason, mid-season, and end-season to determine 0–10, 10–20, and 20–40 m sprint velocity ($m \cdot s^{-1}$), agility ('L run') test time (s), vertical jump height (cm), and VO_{2max} ($ml \cdot kg^{-1} \cdot min^{-1}$)
Gil-Rey et al. [23]	Training sessions and games for 9 weeks during the in-season	Total exposure (min), mean weekly exposure (min), total muscular and respiratory sRPE[a], and mean weekly muscular and respiratory sRPE[a] for elite and sub-elite players	Battery of tests administered pre and post monitoring period to determine time to exhaustion (min), CMJ height (cm), CMJ height with arm swing (cm), 5- and 15-m sprint times (s)
Gorostiaga et al. [28]	Training sessions and games for 45 weeks including two preparatory and two competitive periods	Mean total exposure time (min) between testing sessions and total time (min) spent completing low-, medium-, and high-intensity endurance running and ball exercises based on HR zones calculated using 80% and 90% of HR at [BLa^{-1}] of 3 $mmol \cdot L^{-1}$	Battery of tests administered at the start of the first preparatory period, start and end of the first competitive period, and end of the second competitive period to determine $1RM_{BP}$ (kg), upper-body power (w) at loads of 30, 45, 60, and 70% of 1_{RM}, lower-body power at loads of 60, 80, 100, and 125% of body mass, vertical jump height (cm), 5- and 15-m sprint velocity ($m \cdot s^{-1}$), velocity at [BLa^{-1}] of 3 $mmol \cdot L^{-1}$, and mean standing and running handball-throwing velocity ($m \cdot s^{-1}$)
Granados et al. [29]	Training sessions and games for 45 weeks including two preparatory and two competitive periods	Mean total exposure time between testing sessions	Battery of tests administered at the start of the first preparatory period, start and end of the first competitive period, and end of the second competitive period to determine $1RM_{BP}$ (kg), upper-body power (w) at loads of 30, 45, 60, and 70% of 1_{RM}, lower-body power at loads of 60, 80, 100, and 125% of body mass, vertical jump height (cm), 5- and 15-m sprint velocity ($m \cdot s^{-1}$), velocity at [BLa^{-1}] of 3 $mmol \cdot L^{-1}$, and mean standing and running handball-throwing velocity ($m \cdot s^{-1}$)
Harrison and Johnston [40]	Training sessions for 14 weeks during the preseason	Average individual training load using sRPE[a] (AU)	2-km time trial administered on four separate occasions during the monitoring period 3–4 weeks apart to determine improvements between tests(s)
Horta et al. [41]	Training sessions for 6 weeks during the preseason	Total weekly training load for weeks 2, 4, and 6of the monitoring period using sRPE[a] (AU)	CMJ tests administered every 2 weeks to determine maximal CMJ height (cm)
Los Arcos et al. [24]	Training sessions and games for 9 weeks during the in-season	Total exposure and mean weekly training load using respiratory and muscular sRPE[a]	Battery of tests administered pre and post monitoring period to determine CMJ height (cm), 5- and 15-m sprint times (s), and [BLa^{-1}] 10 and 15 min into the endurance running test

Table 6 (continued)

Study	Monitoring period	Training load measures	Performance measures
Los Arcos et al. [25]	Training sessions and practice games for 5 weeks during the preseason and 4 weeks during the in-season	Total exposure (min) and total respiratory and muscular sRPE[a] (AU)	Battery of tests administered pre and post monitoring period to determine velocity at [BLa^{-1}] of 3 mmol·L^{-1} (km·h^{-1}), [BLa^{-1}] at 12 km·h^{-1} (mmol·L^{-1}), [BLa^{-1}] at 13 km·h^{-1} (mmol·L^{-1}), CMJ height (cm), CMJ height with arm swing (cm), CMJ height on dominant limb (cm), CMJ height on nondominant limb (cm), and 5- and 15-m sprint times (s)
Los Arcos et al. [26]	Training sessions and games for 5 weeks during the preseason and 27 weeks during the in-season	Total exposure (min) and total training load using respiratory and muscular sRPE[a] (AU)	Battery of tests administered pre and post monitoring period to determine CMJ height (cm), CMJ height with arm swing (cm), 5- and 15-m sprint times (s), velocity at [BLa^{-1}] of 3 mmol·L^{-1} (km·h^{-1}), and [BLa^{-1}] at 12 km·h^{-1} and 13 km·h^{-1} (mmol·L^{-1})
Malone et al. [34]	Training sessions for 8 weeks during the in-season	Mean weekly training load using iTRIMP[a] (AU)	Yo-Yo IRT (levels 1 and 2) administered pre and post monitoring period to determine speed at [BLa^{-1}] of 2 mmol·L^{-1}, speed at [BLa^{-1}] of 4 mmol·L^{-1} (km·h^{-1}), running economy (ml·kg^{-1}·min^{-1}), distance travelled (m), and VO_{2max} (ml·kg^{-1}·min^{-1})
Malone et al. [33]	Training sessions for 12 weeks during the in-season	Total time (min) completing low-, medium-, and high-intensity activity based on HR zones calculated using [BLa^{-1}] of 2 and 4 mmol·L^{-1}	Yo-Yo IRT (level 1) administered pre and post monitoring period to determine speed at [BLa^{-1}] of 2 mmol·L^{-1} (km·h^{-1}), speed at [BLa^{-1}] of 4 mmol·L^{-1} (km·h^{-1}), distance travelled (m), and VO_{2max} (ml·kg^{-1}·min^{-1})
Malone et al. [36]	Training sessions and games for 12 weeks during the in-season	Mean weekly training load using sRPE[a], Banister's TRIMP[a], Edward's TRIMP[a], Lucia's TRIMP[a], Team TRIMP[a], and iTRIMP[a] (AU)	Battery of tests administered pre and post monitoring period to determine velocity at LT (km·h^{-1}), velocity at OBLA (km·h^{-1}), VO_{2max} (ml·kg^{-1}·min^{-1}), velocity at VO_{2max} (km·h^{-1}), maximal aerobic speed (km·h^{-1}), running economy (ml·kg^{-1}·min^{-1}), Yo-Yo IRT (level 1 and 2) performance (m), best sprint time during a 35-m repeat-sprint ability test (s), and 5-, 10-, and 20-m sprint times(s)
Manzi et al. [35]	Training sessions for 8 weeks during the preseason	Mean weekly training load using iTRIMP[a] (AU)	Maximal treadmill test administered pre and post monitoring period to determine speed at [BLa^{-1}] of 4 mmol·L^{-1} (km·h^{-1}), VT (ml·kg^{-1}·min^{-1}), VO_2 at VT, and VO_{2max} (ml·kg^{-1}·min^{-1}). Yo-Yo IRT (level 1) administered pre and post monitoring period to determine distance travelled (m)

Table 6 (continued)

Study	Monitoring period	Training load measures	Performance measures
McCaskie et al. [30]	Training sessions for 17 weeks separated into the first 6 weeks and last 11 weeks of the preseason	Total exposure (min), total on-field exposure (min), total running (on-field activities ≥ 1.7 m·s^{-1}) exposure (min), total off-field exposure (min), total training load using total distance (m), run distance (≥ 1.7 m·s^{-1}) (m), high-speed running distance (≥ 5.5 m·s^{-1}) (m), sprint distance (≥ 7.0 m·s^{-1}) (m), PlayerLoad[a], total on-field sRPE[a], total off-field sRPE[a], total sRPE[a] reported for the entire preseason and last 11 weeks of the preseason	Champion Data© player rank during the entire season, first half of the season, and first four games of the season for all players and midfielders/nomadic players only
Nakamura et al. [38]	Training sessions for 9 weeks during the preseason	Total training load and training load for the first 2 weeks of the preseason using sRPE[a] (AU)	Battery of tests administered pre and post monitoring period to determine squat jump height (cm), CMJ height (cm), mean propulsive power relative to body mass during jump squats (W·kg^{-1}), and 5-, 10-, 15-, and 20-m velocity (m·s^{-1})
Stagno et al. [11]	Training sessions and games for 8 weeks during the in-season	Mean weekly training load and mean weekly load for training and games using Team TRIMP[a] (AU) and mean weekly volume of high-intensity activity (> 85% of HR_{max}) (min)	Treadmill tests administered pre and post monitoring period to determine velocity at OBLA (km·h^{-1}), peak running speed (km·h^{-1}), time to exhaustion (s), and VO_{2max} (ml·kg^{-1}·min^{-1})
Taylor et al. [9]	Training sessions and games for 6 weeks during the in-season	Mean weekly training load using total distance (m), individualized high-speed distance (speed ≥ velocity at OBLA) (m), high-speed distance (≥ 15 km·h^{-1}) (m), very high-speed distance (≥ 18 km·h^{-1}) (m), PlayerLoad[a] (AU), iTRIMP[a] (AU), Banister's TRIMP[a] (AU), Lucia's TRIMP[a] (AU), Edwards' TRIMP[a] (AU), and sRPE[a] (AU)	Incremental treadmill test administered pre and post monitoring period to determine VO_{2max} (ml·kg^{-1}·min^{-1}), velocity at VO_{2max} (km·h^{-1}), velocity at OBLA (km·h^{-1}), and velocity at LT (km·h^{-1})

[BLa^{-1}] blood lactate concentration, *1RM* one-repetition maximum, *$1RM_{BP}$* one-repetition maximum bench press, *AD* load acceleration and deceleration distance *AU* arbitrary units, *CMJ* countermovement jump, *COD* change of direction, *HR* heart rate, *HR_{max}* maximum heart rate, *HSD* high-speed distance, *IFT* intermittent fitness test, *IRT* intermittent recovery test, *ISRT* intermittent shuttle run test, *iTRIMP* individualized training impulse, *LT* lactate threshold, *m > 30ASR* distance covered above 30% of anaerobic speed reserve, *m > MAS* distance covered above maximal aerobic speed, *OBLA* onset of blood lactate accumulation, *SD* standard deviation, *sRPE* session rating of perceived exertion, *t > 30ASR* time spent above 30% of anaerobic speed reserve, *t > MAS* time spent above maximal aerobic speed, *TRIMP* training impulse, *VHSD* very high-speed distance, *VO_2* oxygen uptake, *VO_{2max}* maximal oxygen uptake, *VT* ventilatory threshold, *W* watts, *W·kg^{-1}* watts per kilogram

[a]Formula is provided in Table 3

Table 7 Associations between external training load and performance in team sports

Study	Measures	Associations between training load and performance
Brink et al. [22]	1-week exposure vs. ISRT performance (test 2)	SE = 1.68, $P < 0.05$*
	1-week exposure vs. ISRT performance (test 3)	SE = 1.82, $P > 0.05$
	1-week exposure vs. ISRT performance (test 4)	SE = 2.05, $P > 0.05$
	1-week exposure vs. ISRT performance (test 5)	SE = 1.86, $P < 0.05$*
	1-week exposure vs. ISRT performance (test 6)	SE = 1.61, $P < 0.05$*
	1-week exposure vs. ISRT performance (test 7)	SE = 0.33, $P < 0.05$*
	2-week exposure vs. ISRT performance (test 2)	SE = 1.67, $P < 0.05$*
	2-week exposure vs. ISRT performance (test 3)	SE = 1.81, $P > 0.05$
	2-week exposure vs. ISRT performance (test 4)	SE = 1.53, $P < 0.05$*
	2-week exposure vs. ISRT performance (test 5)	SE = 1.75, $P < 0.05$*
	2-week exposure vs. ISRT performance (test 6)	SE = 1.68, $P < 0.05$*
	2-week exposure vs. ISRT performance (test 7)	SE = 1.84, $P < 0.05$*
Campos-Vazquez et al. [7]	Training exposure vs. IFT performance	Unclear
	Friendly match exposure vs. IFT performance	$r = 0.68$, large
Caparros et al. [8]	Total number of practices vs. total team ranking[a]	$r = 0.67$, $P = 0.09$
	Total number of practices vs. mean game ranking[a]	$r = 0.14$, $P = 0.75$
	Total number of practices vs. mean player ranking[a]	$r = 0.52$, $P = 0.23$
	Total number of practices vs. mean player points scored	$r = 0.77$, $P = 0.04$*
	Total number of games vs. total team ranking[a]	$r = 0.87$, $P = 0.01$*
	Total number of games vs. mean game ranking[a]	$r = 0.39$, $P = 0.39$
	Total number of games vs. mean player ranking[a]	$r = 0.57$, $P = 0.18$
	Total number of games vs. mean player points scored	$r = 0.71$, $P = 0.74$
	Total hours exposure vs. total team ranking[a]	$r = 0.76$, $P = 0.01$*
	Total hours exposure vs. mean game ranking[a]	$r = 0.35$, $P = 0.44$
	Total hours exposure vs. mean player ranking[a]	$r = 0.48$, $P = 0.27$
	Total hours exposure vs. mean player points scored	$r = 0.70$, $P = 0.77$
	Total hours practice exposure vs. total team ranking[a]	$r = 0.80$, $P = 0.03$*
	Total hours practice exposure vs. mean game ranking[a]	$r = 0.23$, $P = 0.61$
	Total hours practice exposure vs. mean player ranking[a]	$r = 0.51$, $P = 0.24$
	Total hours practice exposure vs. mean player points scored	$r = 0.43$, $P = 0.34$
	Total hours game exposure vs. total team ranking[a]	$r = 0.75$, $P = 0.54$
	Total hours game exposure vs. mean game ranking[a]	$r = 0.35$, $P = 0.43$
	Total hours game exposure vs. mean player ranking[a]	$r = 0.48$, $P = 0.28$
	Total hours game exposure vs. mean player scored points	$r = 0.71$, $P = 0.07$
Fitzpatrick et al. [27]	Total distance vs. maximal sprint speed	$r = 0.46$, moderate, $R^2 = 0.21$
	Total distance vs. maximal aerobic speed	$r = 0.26$, unclear, R^2 = unclear
	AD load vs. maximal sprint speed	$r = 0.57$, large, $R^2 = 0.32$
	AD load vs. maximal aerobic speed	$r = 0.20$, unclear, R^2 = unclear
	m > HSD vs. maximal sprint speed	$r = 0.32$, unclear, $R^2 = 0.11$
	m > HSD vs. maximal aerobic speed	$r = 0.22$, unclear, $R^2 = 0.05$
	t > HSD vs. maximal sprint speed	$r = 0.34$, unclear, $R^2 = 0.11$
	t > HSD vs. maximal aerobic speed	$r = 0.37$, unclear, $R^2 = 0.14$
	m > MAS vs. maximal sprint speed	$r = 0.30$, unclear, $R^2 = 0.09$
	m > MAS vs. maximal aerobic speed	$r = 0.50$, large, $R^2 = 0.25$
	t > MAS vs. maximal sprint speed	$r = 0.21$, unclear, $R^2 = 0.04$
	t > MAS vs. maximal aerobic speed	$r = 0.77$, very large, $R^2 = 0.59$

Table 7 (continued)

Study	Measures	Associations between training load and performance
	m > VHSD vs. maximal sprint speed	$r = 0.25$, unclear, $R^2 = 0.06$
	m > VHSD vs. maximal aerobic speed	$r = -0.07$, unclear, $R^2 = 0.00$
	t > VHSD vs. maximal sprint speed	$r = 0.34$, unclear, $R^2 = 0.07$
	t > VHSD vs. maximal aerobic speed	$r = 0.37$, unclear, $R^2 = 0.00$
	m > 30ASR vs. maximal sprint speed	$r = -0.09$, unclear, $R^2 = 0.01$
	m > 30ASR vs. maximal aerobic speed	$r = 0.20$, unclear, $R^2 = 0.04$
	t > 30ASR vs. maximal sprint seed	$r = -0.15$, unclear, $R^2 = 0.02$
	t > 30ASR vs. maximal aerobic speed	$r = 0.62$, large, $R^2 = 0.38$
Gil-Rey et al. [23]	Total exposure vs. time to exhaustion	$r = 0.67$, large
	Total exposure vs. CMJ height	$r = -0.16$, small
	Total exposure vs. CMJ height with arm swing	$r = 0.10$, small
	Total exposure vs. 5-m sprint time	$r = -0.04$, trivial
	Total exposure vs. 15-m sprint time	$r = 0.23$, small
Gorostiaga et al. [28]	Test 1 – Test 2: Strength training exposure vs. standing handball-throwing velocity	$r = 0.58$, $P < 0.05$*
Granados et al. 2008 [29]	Test 1 – Test 2: Practice and competitive game exposure vs. upper-body power at 30% of $1RM_{BP}$	$r = 0.62$, $P < 0.05$*
Los Arcos et al. [24]	Total exposure vs. CMJ height	$r = -0.62$, large, $P < 0.01$*
Los Arcos et al. [25]	Total exposure vs. velocity at [BLa^{-1}] of 3 mmol·L^{-1}	$r = -0.31$, moderate, $P > 0.05$
	Total exposure vs. [BLa^{-1}] at 12 km·h^{-1}	$r = -0.21$, small, $P > 0.05$
	Total exposure vs. [BLa^{-1}] at 13 km·h^{-1}	$r = -0.37$, moderate, $P > 0.05$
	Total exposure vs. CMJ height	$r = -0.42$, moderate, $P > 0.05$
	Total exposure vs. CMJ height with arm swing	$r = -0.51$, large, $P < 0.05$*
	Total exposure vs. CMJ height (dominant limb)	$r = -0.27$, small, $P > 0.05$
	Total exposure vs. CMJ height (nondominant limb)	$r = -0.39$, moderate, $P > 0.05$
	Total exposure vs. 5-m sprint time	$r = -0.54$, large, $P < 0.05$*
	Total exposure vs. 15-m sprint time	$r = -0.64$, large, $P < 0.05$*
Los Arcos et al. [26]	Total exposure vs. CMJ height	$r = 0.36$, unclear
	Total exposure vs. CMJ height with arm swing	$r = 0.32$, unclear
	Total exposure vs. 5-m sprint time	$r = 0.05$, unclear
	Total exposure vs. 15-m sprint time	$r = -0.23$, unclear
	Total exposure vs. velocity at [BLa^{-1}] of 3 mmol·L^{-1}	$r = -0.57$, large
	Total exposure vs. [BLa^{-1}] at 12 km·h^{-1}	$r = -0.62$, large
	Total exposure vs. [BLa^{-1}] at 13 km·h^{-1}	$r = -0.61$, large
McCaskie et al. [30]	*Entire preseason (all player):*	
	Total exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (first four games)	$r = 0.33$, moderate, $P = 0.04$*
	Total off-field exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total distance vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total distance vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total distance vs. Champion Data© player rank[a] (first four games)	$P > 0.05$

Table 7 (continued)

Study	Measures	Associations between training load and performance
	Total run distance vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total run distance vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total run distance vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total HSD vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total HSD vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total HSD vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Last 11 weeks of the preseason (all players):	
	Total exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank (first half of season)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (first four games)	$r = 0.50$, moderate, $P = 0.001$*
	Total off-field exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first four games)	$r = 0.33$, moderate, $P = 0.04$*
	Total distance vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total distance vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total distance vs. Champion Data© player rank[a] (first four games)	$r = 0.37$, moderate, $P = 0.02$*
	Total run distance vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total run distance vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total run distance vs. Champion Data© player rank[a] (first four games)	$r = 0.33$, moderate, $P = 0.03$*
	Total HSD vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total HSD vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total HSD vs. Champion Data© player rank[a] (first four games)	$r = 0.36$, moderate, $P = 0.02$*
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first four games)	$r = 0.38$, moderate, $P = 0.02$*
	Entire preseason (midfielders/nomadic players only):	
	Total exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first four games)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total running exposure vs. Champion Data© player rank[a] (first four games)	$r = 0.44$, moderate, $P = 0.06$
	Total off-field exposure vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first four games)	$r = -0.31$, moderate, $P = 0.19$
	Total distance vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total distance vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$

Table 7 (continued)

Study	Measures	Associations between training load and performance
	Total distance vs. Champion Data© player rank[a] (first four games)	$r=0.31$, moderate, $P=0.31$
	Total run distance vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total run distance vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total run distance vs. Champion Data© player rank[a] (first four games)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (first four games)	$r=0.36$, moderate, $P=0.12$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first four games)	$r=0.37$, moderate, $P=0.11$
	Last 11 weeks of the preseason (midfielders/nomadic players only):	
	Total exposure vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total exposure vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total exposure vs. Champion Data© player rank[a] (first four games)	$P>0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total on-field exposure vs. Champion Data© player rank[a] (first four games)	$P>0.05$
	Total running exposure vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total running exposure vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total running exposure vs. Champion Data© player rank[a] (first four games)	$r=0.41$, moderate, $P=0.07$
	Total off-field exposure vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total off-field exposure vs. Champion Data© player rank[a] (first four games)	$r=-0.37$, moderate, $P=0.11$
	Total distance vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total distance vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total distance vs. Champion Data© player rank[a] (first four games)	$P>0.05$
	Total run distance vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total run distance vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (first four games)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total HSD vs. Champion Data© player rank[a] (first four games)	$r=0.32$, moderate, $P=0.17$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (entire season)	$P>0.05$
	Total PlayerLoad[a] vs. Champion Data© player rank[a] (first half of season)	$P>0.05$
	Total PlayerLoad[a] vs. Champion Data© total player rank[a] (first four games)	$r=0.34$, moderate, $P=0.15$
Taylor et al. [9]	Total distance vs. velocity at LT	$r=-0.21$, small, $P=0.56$
	Total distance vs. velocity at OBLA	r=-0.31, moderate, $P=0.38$
	Total distance vs. velocity at VO_{2max}	$r=-0.002$, trivial, $P=0.99$
	Total distance vs. VO_{2max}	$r=-0.51$, large, $P=0.13$
	Individualized HSD vs. velocity at LT	$r=0.12$, small, $P=0.75$
	Individualized HSD vs. velocity at OBLA	$r=0.27$, small, $P=0.46$
	Individualized HSD vs. velocity at VO_{2max}	$r=0.34$, moderate, $P=0.33$
	Individualized HSD vs. VO_{2max}	$r=-0.26$, small, $P=0.47$
	HSD vs. velocity at LT	$r=-0.06$, trivial, $P=0.87$
	HSD vs. velocity at OBLA	$r=0.25$, small, $P=0.49$
	HSD vs. velocity at VO_{2max}	$r=0.32$, moderate, $P=0.36$
	HSD vs. VO_{2max}	$r=-0.19$, small, $P=0.59$

Table 7 (continued)

Study	Measures	Associations between training load and performance
	VHSD vs. velocity at LT	$r = -0.43$, moderate, $P = 0.22$
	VHSD vs. velocity at OBLA	$r = -0.66$, large, $P = 0.04$
	VHSD vs. velocity at VO_{2max}	$r = -0.16$, small, $P = 0.66$
	VHSD vs. VO_{2max}	$r = -0.63$, large, $P = 0.05$
	PlayerLoad[a] vs. velocity at LT	$r = -0.03$, trivial, $P = 0.93$
	PlayerLoad[a] vs. velocity at OBLA	$r = -0.47$, moderate, $P = 0.17$
	PlayerLoad[a] vs. velocity at VO_{2max}	$r = 0.17$, small, $P = 0.64$
	PlayerLoad[a] vs. VO_{2max}	$r = -0.24$, small, $P = 0.50$

[BLa⁻¹] blood lactate concentration, $1RM_{BP}$ one-repetition maximum bench press, *AD* load acceleration and deceleration distance > 2 m·s^{-2}, *CMJ* countermovement jump, *HSD* high-speed distance, *IFT* intermittent fitness test, *ISRT* intermittent shuttle run test, *LT* lactate threshold, *m > 30ASR* distance covered above 30% of anaerobic speed reserve, *m > HSD* distance covered above 17 km·h^{-1}, *m > MAS* distance covered above maximal aerobic speed, *m > VHSD* distance covered above 21 km·h^{-1}, *OBLA* onset of blood lactate accumulation, *r* correlation coefficient, *SE* standard error, *t > 30ASR* time spent above 30% of anaerobic speed reserve, *t > HSD* time spent above 17 km·h^{-1}, *t > MAS* time spent above maximal aerobic speed, *t > VHSD* time spent above 21 km·h^{-1}, *VHSD* very high-speed distance, VO_{2max} maximal oxygen uptake

*Indicates a statistically significant association between variables

[a]Formula is provided in Table 3

before each test could predict intermittent shuttle run test performance. They found that 1-week exposure predicted performance in four of the six intermittent shuttle run tests analyzed (test one was used to determine baseline performance); however, 2-week exposure predicted performance in five of the six tests analyzed [22].

Four studies [23–26] examined the correlation between total exposure (training and games) and changes in aerobic performance. All correlations were nonsignificant or statistical significance was not reported, and magnitudes ranged from small to large [23–26]. The associations between other external load variables and changes in aerobic performance were only assessed in two studies [9, 27]. Taylor et al. [9] examined correlations between total distance, individualized high-speed distance, high-speed distance, very high-speed distance, PlayerLoad™, and changes in aerobic performance. All correlations were nonsignificant and ranged from trivial to large [9]. Fitzpatrick et al. [27] conducted correlations and linear regression analyses between total distance, acceleration and deceleration load, distance covered above 17 km·h^{-1}, duration spent working above 17 km·h^{-1}, distance covered above 21 km·h^{-1}, duration spent working above 21 km·h^{-1}, distance covered above maximal aerobic speed, duration spent working above maximal aerobic speed, distance covered above 30% of anaerobic speed reserve, duration spent working above 30% of anaerobic speed reserve, and changes in aerobic reserve. Correlations ranged from unclear to very large, coefficient of determination (R^2) ranged from unclear to 0.59, and statistical significance was not reported [27].

Correlations between exposure to individual training modes and changes in aerobic performance were assessed in one study [7]. Campos-Vazquez et al. [7] investigated correlations between training exposure and practice game exposure with changes in intermittent fitness test (IFT) performance. The correlation between training exposure and IFT performance was unclear (confidence limits overlapped positive and negative values), whereas the relationship between practice game exposure and IFT performance was large and statistical significance was not reported [7].

3.5.2 External Training Load and Neuromuscular Variables

Four studies [23–26] examined correlations between total exposure and changes in neuromuscular variables. Los Arcos et al. [25] reported significant, large correlations between total exposure, countermovement jump (CMJ) height with arm swing, and 5- and 15-m sprint time, whereas correlations with other neuromuscular variables (CMJ height without arm swing, CMJ height on dominant limb, and CMJ height on nondominant limb) were nonsignificant and ranged from small to moderate. In another study, Los Arcos et al. [24] reported a significant, large correlation between total exposure and CMJ height. In contrast, Gil-Rey et al. [23] reported trivial to small correlations between total exposure and neuromuscular variables (CMJ height with and without arm swing and 5- and 15-m sprint times); statistical significance was not determined. Similarly, Los Arcos et al. [26] reported unclear correlations between total exposure and changes in neuromuscular variables (CMJ height, CMJ

height with arm swing, and 5- and 15-m sprint times); statistical significance was not reported.

Correlations between exposure to individual training modes and changes in neuromuscular variables were assessed in two studies [28, 29]. Gorostiaga et al. [28] reported a significant correlation between strength training exposure and handball-throwing velocity, whereas all other correlations between exposure and neuromuscular variables (one-repetition maximum bench press [$1RM_{BP}$]; upper-body power during bench press at loads of 30, 45, 60, and 70% of $1RM_{BP}$; lower-body power during squats at loads of 60, 80, 100, and 125% of body mass; vertical jump height; and 5- and 15-m sprint speeds) were nonsignificant, and no interpretations were reported. In another study, Granados et al. [29] reported a significant association between exposure to practice games and competitive games combined and upper-body power during bench press exercise at a load of 30% of $1RM_{BP}$, whereas all other associations between exposure and neuromuscular variables ($1RM_{BP}$; upper-body power during bench press at loads of 45, 60, and 70% of 1_{RM}; lower-body power during squats at loads of 60, 80, 100, and 125% of body mass; vertical jump height; 5- and 15-m sprint speeds; and handball-throwing velocity) were nonsignificant, and no interpretations were reported.

3.5.3 External Training Load and Game-Related Statistics

Correlations between exposure and game-related statistics were investigated in two studies [8, 30]. Caparros et al. [8] monitored basketball players to determine the correlations between training sessions and games, training hours and game hours, and total exposure hours per season over seven seasons, with total team ranking, mean game ranking, mean player ranking, and mean player points scored (see formulae in Table 3). The correlation between the number of training sessions and mean player points scored was significant, but no interpretations were provided [8]. The correlations between the number of games, training hours, total exposure hours, and total team ranking were significant; no interpretations were provided [8]. All other correlations were nonsignificant, and no interpretations were provided [8].

McCaskie et al. [30] investigated the association between exposure, PlayerLoad™, and a range of global positioning system (GPS)-derived variables (see Table 6) (last 11 weeks and entire preseason) and the Champion Data© player rank (first four games, first half, and entire season). For all players, total running exposure during the entire preseason and running exposure, off-field exposure, total distance, running distance, high-speed running distance, and total PlayerLoad™ during the last 11 weeks were significantly and moderately associated with Champion Data© player rank for the first four games of the season [30]. All other correlations were nonsignificant [30].

3.6 Associations Between Internal Training Load and Performance

3.6.1 Intensity Distribution and Aerobic Performance

Associations between internal training load and performance are presented in Table 8. In five studies [11, 28, 31–33], internal training load was reported as the total time spent completing low-, medium-, and high-intensity activity using zones based on proportions of maximum HR. Correlations between time spent performing high-intensity activity and changes in aerobic performance were significant in all studies [11, 28, 31–33], and interpretations reported ranged from large to very large. Nonsignificant associations between high-intensity activity and aerobic performance (time to exhaustion and peak running speed during the maximal treadmill test) were also reported in one study; no interpretation was provided [11].

In two studies [7, 28], correlations between intensity distribution during individual training modes and changes in aerobic performance were assessed separately. Campos-Vazquez et al. [7] reported very large correlations between high-intensity activity during practice games and aerobic performance and unclear correlations between the volume of high-intensity activity during training sessions and aerobic performance; statistical significance was not reported. Gorostiaga et al. [28] reported the volume of low-, medium-, and high-intensity endurance running and ball exercises separately, reporting significant correlations between time spent performing medium- and high-intensity endurance activity combined and velocity at a blood lactate concentration [BLa^{-1}] of 3 $mmol \cdot L^{-1}$; no interpretations were provided.

3.6.2 Intensity Distribution and Neuromuscular Variables

Correlations between intensity distribution and changes in neuromuscular variables were assessed in one study [28]. Gorostiaga et al. [28] determined whether time spent working at low, medium, and high intensities were correlated with changes in neuromuscular variables. The volume of low-intensity activity was significantly correlated with lower-body power during a squat at a load of 125% of body mass between tests 1 and 2, and between tests 3 and 4, but there were no significant correlations with other neuromuscular variables ($1RM_{BP}$; upper-body power during bench press at loads of 30, 45, 60, and 70% of $1RM_{BP}$; lower-body power during squats at loads of 60, 80, and 100% of body mass; vertical jump height; 5- and 15-m sprint velocity; and handball-throwing velocity); no interpretations were provided.

Table 8 Associations between internal training load and performance in team sports

Study	Measures	Associations between training load and performance
Akubat et al. [3]	sRPE[a] vs. velocity at LT	$r = 0.13$, small, $P > 0.05$
	sRPE[a] vs. HR at LT	$r = 0.20$, small, $P > 0.05$
	sRPE[a] vs. velocity at OBLA	$r = 0.40$, moderate, $P > 0.05$
	sRPE[a] vs. HR at OBLA	$r = 0.15$, small, $P > 0.05$
	Banister's TRIMP[a] vs. velocity at LT	$r = 0.28$, small, $P > 0.05$
	Banister's TRIMP[a] vs. HR at LT	$r = 0.21$, small, $P > 0.05$
	Banister's TRIMP[a] vs. velocity at OBLA	$r = 0.43$, moderate, $P > 0.05$
	Banister's TRIMP[a] vs. HR at OBLA	$r = -0.21$, small, $P > 0.05$
	Team TRIMP[a] vs. velocity at LT	$r = 0.20$, small, $P > 0.05$
	Team TRIMP[a] vs. HR at LT	$r = 0.28$, small, $P > 0.05$
	Team TRIMP[a] vs. velocity at OBLA	$r = 0.28$, small, $P > 0.05$
	Team TRIMP[a] vs. HR at OBLA	$r = -0.49$, moderate, $P > 0.05$
	iTRIMP[a] vs. velocity at LT	$r = 0.67$, large, $P < 0.05$*
	iTRIMP[a] vs. HR at LT	$r = 0.17$, small, $P > 0.05$
	iTRIMP[a] vs. velocity at OBLA	$r = 0.33$, moderate, $P > 0.05$
	iTRIMP[a] vs. HR at OBLA	$r = -0.25$, small, $P > 0.05$
Aughey et al. [42]	sRPE[a] vs. game outcome	ES = 0.43, moderate
	sRPE[a] vs. game outcome with covariate of days brea	ES = 0.45, moderate
	sRPE[a] vs. game outcome with covariate of ladder position	ES = 0.29, moderate
	Training–stress balance[a] (sRPE) vs. game outcome	ES = 0.31, moderate
	Training–stress balance[a] (sRPE) vs. game outcome with covariate of days break	ES = 0.39, moderate
	Training–stress balance[a] (sRPE) vs. game outcome with covariate of ladder position	ES = 0.27, moderate
	Training–stress balance[a] (training strain[a]) vs. game outcome	ES = 0.51, moderate
	Training–stress balance[a] (training strain[a]) vs. game outcome with covariate of days break	ES = 0.48, moderate
	Training–stress balance[a] (training strain[a]) vs. game outcome with covariate of ladder position	ES = − 0.02, small
Campos-Vazquez et al. [7]	Total sRPE[a] vs. maximal velocity during the IFT	$r = 0.70$, very large
	Total Edwards' TRIMP[a] vs. maximal velocity during the IFT	$r = 0.25$, moderate
	Edwards' TRIMP[a] during practice games vs. maximal velocity during the IFT	$r = 0.66$, very large
	Edwards' TRIMP[a] during training sessions vs. maximal velocity during the IFT	Unclear
	Volume of high-intensity activity during practice games vs. maximal velocity during the IFT	$r = 0.68$, very large
	Volume of high-intensity activity during training sessions vs. maximal velocity during the IFT	Unclear
Castagna et al. [31]	Volume of high-intensity activity vs. speed at [BLa^{-1}] of 2 mmol·L^{-1}	$r = 0.84$, very large, $P < 0.001$*
	Volume of high-intensity activity vs. speed at [BLa^{-1}] of 4 mmol·L^{-1}	$r = 0.65$, large, $P < 0.01$*
Castagna et al. [31]	Volume of high-intensity activity vs. speed at [BLa^{-1}] of 2 mmol·L^{-1}	$r = 0.78$, very large, $P = 0.002$*
	Volume of high-intensity activity vs. speed at [BLa^{-1}] of 4 mmol·L^{-1}	$r = 0.60$, large, $P = 0.03$*
	Volume of high-intensity activity vs. VO_{2max}	$r = 0.65$, large, $P = 0.02$*
	Volume of high-intensity activity vs. Yo-Yo IRT (level 1) performance	$r = 0.66$, large, $P = 0.01$*
Dobbin et al. [39]	Resistance training sRPE[a] vs. 10-m sprint time	$r = -0.52$, moderate, $R^2 = 0.27$
	Resistance training sRPE[a] vs. 20-m sprint time	$r = -0.49$, small, $R^2 = 0.24$
	Resistance training sRPE[a] vs. 10-m momentum	$r = 0.12$, $R^2 = 0.01$
	Resistance training sRPE[a] vs. 20-m momentum	$r = 0.01$, $R^2 = 0.00$

Table 8 (continued)

Study	Measures	Associations between training load and performance
	Resistance training sRPE[a] vs. CMJ height	$r=0.51$, moderate, $R^2=0.26$
	Resistance training sRPE[a] vs. COD test time	$r=0.42$, small, $R^2=0.18$
	Resistance training sRPE[a] vs. medicine ball throwing distance	$r=0.40$, small, $R^2=0.16$
	Resistance training sRPE[a] vs. Yo-Yo IR1 distance	$r=0.04$, $R^2=0.00$
	Conditioning training sRPE[a] vs. 10-m sprint time	$r=0.47$, small, $R^2=0.22$
	Conditioning training sRPE[a] vs. 20-m sprint time	$r=0.65$, moderate, $R^2=0.42$
	Conditioning training sRPE[a] vs. 10-m momentum	$r=0.51$, moderate, $R^2=0.26$
	Conditioning training sRPE[a] vs. 20-m momentum	$r=0.53$, moderate, $R^2=0.28$
	Conditioning training sRPE[a] vs. CMJ height	$r=0.19$, $R^2=0.04$
	Conditioning training sRPE[a] vs. COD test time	$r=0.14$, $R^2=0.02$
	Conditioning training sRPE[a] vs. medicine ball throwing distance	$r=0.03$, $R^2=0.00$
	Conditioning training sRPE[a] vs. Yo-Yo IR1 distance	$r=0.01$, $R^2=0.00$
	Skills training sRPE[a] vs. 10-m sprint time	$r=0.71$, large, $R^2=0.51$
	Skills training sRPE[a] vs. 20-m sprint time	$r=0.79$, large, $R^2=0.62$
	Skills training sRPE[a] vs. 10-m momentum	$r=0.35$, small, $R^2=0.12$
	Skills training sRPE[a] vs. 20-m momentum	$r=0.27$, $R^2=0.07$
	Skills training sRPE[a] vs. CMJ height	$r=0.60$, moderate, $R^2=0.36$
	Skills training sRPE[a] vs. COD test time	$r=0.20$, $R^2=0.04$
	Skills training sRPE[a] vs. medicine ball throwing distance	$r=0.22$, $R^2=0.05$
	Skills training sRPE[a] vs. Yo-Yo IR1 distance	$r=0.11$, $R^2=0.01$
	Total sRPE[a] vs. 10-m sprint time	$r=-0.70$, moderate, $R^2=0.49$
	Total sRPE[a] vs. 20-m sprint time	$r=-0.77$, large, $R^2=0.59$
	Total sRPE[a] vs. 10-m momentum	$r=0.36$, small, $R^2=0.13$
	Total sRPE[a] vs. 20-m momentum	$r=0.29$, $R^2=0.08$
	Total sRPE[a] vs. CMJ height	$r=0.55$, moderate, $R^2=0.31$
	Total sRPE[a] vs. COD test time	$r=0.32$, $R^2=0.11$
	Total sRPE[a] vs. medicine ball throwing distance	$r=0.29$, $R^2=0.08$
	Total sRPE[a] vs. Yo-Yo IR1 distance	$r=0.07$, $R_2=0.01$
Gabbett and Domrow [37]	*Preseason:*	
	sRPE[a] vs. 10-m sprint speed	$P=0.39$
	sRPE[a] vs. 20-m sprint speed	$P=0.53$
	sRPE[a] vs. 40-m sprint speed	$P=0.45$
	sRPE[a] vs. agility test time	$P=0.82$
	sRPE[a] vs. vertical jump height	$P=0.57$
	sRPE[a] vs. VO_{2max}	$P=0.20$
	Early competition:	
	sRPE[a] vs. 10-m sprint speed	$P=0.77$
	sRPE[a] vs. 20-m sprint speed	$P=0.41$
	sRPE[a] vs. 40-m sprint speed	$P=0.16$
	sRPE[a] vs. agility test time	$P=0.04$*
	sRPE[a] vs. vertical jump height	$P=0.90$
	sRPE[a] vs. VO_{2max}	$P=0.31$
	Late competition:	
	sRPE[a] vs. 10-m sprint speed	$P=0.47$
	sRPE[a] vs. 20-m sprint speed	$P=0.73$
	sRPE[a] vs. 40-m sprint speed	$P=0.58$
	sRPE[a] vs. agility test time	$P=0.83$
	sRPE[a] vs. vertical jump height	$P=0.99$
	sRPE[a] vs. VO_{2max}	$P=0.66$

Table 8 (continued)

Study	Measures	Associations between training load and performance
Gil-Rey et al. [23]	Respiratory sRPE[a] vs. time to exhaustion	$r = 0.71$, very large
	Respiratory sRPE[a] vs. CMJ height	$r = -0.06$, trivial
	Respiratory sRPE[a] vs. CMJ with arm swing height	$r = 0.25$, small
	Respiratory sRPE[a] vs. 5-m sprint time	$r = -0.02$, trivial
	Respiratory sRPE[a] vs. 15-m sprint time	$r = -0.21$, small
	Muscular sRPE[a] vs. time to exhaustion	$r = 0.69$, large
	Muscular sRPE[a] vs. CMJ height	$r = -0.17$, small
	Muscular sRPE[a] vs. CMJ with arm swing height	$r = 0.17$, small
	Muscular sRPE[a] vs. 5-m sprint time	$r = -0.06$, trivial
	Muscular sRPE[a] vs. 15-m sprint time	$r = -0.15$, small
Gorostiaga et al. [28]	*Test 1—Test 2:*	
	Volume of low-intensity activity vs. lower-body power at a load of 125% of body mass	$r = -0.79$, $P < 0.01$*
	Test 2—Test 3:	
	Volume of high-intensity activity vs. velocity at [BLa^{-1}] of 3 mmol·L^{-1}	$r = 0.68$, $P < 0.05$*
	Test 3—Test 4:	
	Volume of low-intensity activity vs. lower-body power at a load of 125% of body mass	$r = -0.93$, $P < 0.05$*
	Volume of medium- and high-intensity endurance running vs. velocity at [BLa^{-1}] of 3 mmol·L^{-1}	$r = 0.95$, $P < 0.05$*
Fitzpatrick et al. [27]	Edwards' TRIMP[a] vs. maximal sprint speed	$r = 0.40$, moderate
	Edwards' TRIMP[a] vs. maximal aerobic speed	$r = -0.21$, unclear
	sRPE[a] vs. maximal sprint speed	$r = 0.37$, unclear
	sRPE[a] vs. maximal aerobic speed	$r = 0.22$, unclear
Harrison and Johnston [40]	sRPE[a] vs. 2-km time trial performance	$r = 0.43$, $P = 0.002$*
Horta et al. [41]	sRPE[a] vs. CMJ height	$P > 0.05$
Los Arcos et al. [24]	Muscular sRPE vs. 15-m sprint time	$r = -0.59$, large, $P \leq 0.05$*
Los Arcos et al. [25]	Respiratory sRPE[a] vs. CMJ height	$r = -0.44$, moderate, $P > 0.05$
	Respiratory sRPE[a] vs. CMJ with arm swing height	$r = -0.29$, small, $P > 0.05$
	Respiratory sRPE[a] vs. CMJ height on dominant limb	$r = -0.23$, small, $P > 0.05$
	Respiratory sRPE[a] vs. CMJ height on nondominant limb	$r = -0.30$, small, $P > 0.05$
	Respiratory sRPE[a] vs. 5-m sprint time	$r = -0.37$, moderate, $P > 0.05$
	Respiratory sRPE[a] vs. 15-m sprint time	$r = -0.49$, moderate, $P > 0.05$
	Respiratory sRPE[a] vs. velocity at [BLa^{-1}] of 3 mmol·L^{-1}	$r = -0.30$, small, $P > 0.05$
	Respiratory sRPE[a] vs. [BLa^{-1}] at 12 km·h^{-1}	$r = 0.01$, trivial, $P > 0.05$
	Respiratory sRPE[a] vs. [BLa^{-1}] at 13 km·h^{-1}	$r = -0.36$, moderate, $P > 0.05$
	Muscular sRPE[a] vs. CMJ height	$r = -0.20$, small, $P > 0.05$
	Muscular sRPE[a] vs. CMJ with arm swing height	$r = -0.40$, moderate, $P > 0.05$
	Muscular sRPE[a] vs. CMJ height on dominant limb	$r = -0.54$, large, $P < 0.05$*
	Muscular sRPE[a] vs. CMJ height on nondominant limb	$r = -0.52$, large, $P < 0.05$*
	Muscular sRPE[a] vs. 5-m sprint time	$r = 0.00$, trivial, $P > 0.05$
	Muscular sRPE[a] vs. 15-m sprint time	$r = -0.15$, small, $P > 0.05$
	Muscular sRPE[a] vs. velocity at [BLa^{-1}] of 3 mmol·L^{-1}	$r = -0.45$, moderate, $P > 0.05$
	Muscular sRPE[a] vs. [BLa^{-1}] at 12 km·h^{-1}	$r = -0.29$, small, $P > 0.05$
	Muscular sRPE[a] vs. [BLa^{-1}] at 13 km·h^{-1}	$r = -0.57$, large, $P < 0.05$*
Los Arcos et al. [26]	Respiratory sRPE[a] vs. CMJ height	$r = 0.10$, unclear
	Respiratory sRPE[a] vs. CMJ height with arm swing	$r = 0.31$, unclear
	Respiratory sRPE[a] vs. 5-m sprint time	$r = -0.36$, unclear
	Respiratory sRPE[a] vs. 15-m sprint time	$r = -0.53$, large

Table 8 (continued)

Study	Measures	Associations between training load and performance
	Respiratory sRPE[a] vs. velocity at [BLa-1] of 3 mmol·L-1	$r = -0.57$, large
	Respiratory sRPE[a] vs. [BLa^{-1}] at 12 km·h^{-1}	$r = -0.34$, unclear
	Respiratory sRPE[a] vs. [BLa^{-1}] at 13 km·h^{-1}	$r = -0.61$, large
	Muscular sRPE[a] vs. CMJ height	$r = 0.13$, unclear
	Muscular sRPE[a] vs. CMJ height with arm swing	$r = 0.30$, unclear
	Muscular sRPE[a] vs. 5-m sprint time	$r = -0.16$, unclear
	Muscular sRPE[a] vs. 15-m sprint time	$r = -0.34$, unclear
	Muscular sRPE[a] vs. velocity at [BLa-1] of 3 mmol·L-1	$r = -0.47$, moderate
	Muscular sRPE[a] vs. [BLa^{-1}] at 12 km·h^{-1}	$r = -0.14$, unclear
	Muscular sRPE[a] vs. [BLa^{-1}] at 13 km·h^{-1}	$r = -0.55$, large
Malone and Collins [34]	iTRIMP[a] vs. speed at [BLa^{-1}] of 2 mmol·L^{-1}	$r = 0.64$, large, $P = 0.004$*
	iTRIMP[a] vs. speed at [BLa^{-1}] of 4 mmol·L^{-1}	$r = 0.78$, very large, $P = 0.003$*
	iTRIMP[a] vs. running economy	$r = 0.78$, very large, $P = 0.002$*
	iTRIMP[a] vs. VO_{2max}	$r = 0.77$, very large, $P = 0.002$*
	iTRIMP[a] vs. Yo-Yo IRT (level 1) performance	$r = 0.69$, large, $P = 0.009$*
	iTRIMP[a] vs. Yo-Yo IRT (level 2) performance	$r = 0.69$, large, $P = 0.009$*
Malone et al. [33]	Volume of high-intensity activity vs. speed at [BLa^{-1}] of 2 mmol·L^{-1}	$r = 0.80$, very large, $P = 0.04$*
	Volume of high-intensity activity vs. speed at [BLa^{-1}] of 4 mmol·L^{-1}	$r = 0.58$, large, $P = 0.03$*
	Volume of high-intensity activity vs. VO_{2max}	$r = 0.77$, very large, $P = 0.001$*
	Volume of high-intensity activity vs. Yo-Yo IRT (level 1) performance	$r = 0.65$, large, $P = 0.05$*
Malone et al. [36]	sRPE[a] vs. velocity at LT	$r = 0.13$, $P > 0.05$
	sRPE[a] vs. velocity at OBLA	$r = 0.40$, $P > 0.05$
	sRPE[a] vs. VO_{2max}	$r = 0.24$, $P > 0.05$
	sRPE[a] vs. velocity at VO_{2max}	$r = 0.41$, $P > 0.05$
	sRPE[a] vs. maximal aerobic speed	$r = 0.46$, $P > 0.05$
	sRPE[a] vs. running economy	$r = 0.31$, $P > 0.05$
	sRPE[a] vs. Yo-Yo IR1 performance	$r = 0.12$, $P > 0.05$
	sRPE[a] vs. Yo-Yo IR2 performance	$r = 0.11$, $P > 0.05$
	sRPE[a] vs. repeat-sprint test time	$r = 0.17$, $P > 0.05$
	sRPE[a] vs. 5-m sprint time	$r = 0.11$, $P > 0.05$
	sRPE[a] vs. 10-m sprint time	$r = 0.13$, $P > 0.05$
	sRPE[a] vs. 20-m sprint time	$r = 0.14$, $P > 0.05$
	Banister's TRIMP[a] vs. velocity at LT	$r = 0.28$, $P > 0.05$
	Banister's TRIMP[a] vs. velocity at OBLA	$r = 0.44$, $P > 0.05$
	Banister's TRIMP[a] vs. VO_{2max}	$r = 0.31$, $P > 0.05$
	Banister's TRIMP[a] vs. velocity at VO_{2max}	$r = 0.21$, $P > 0.05$
	Banister's TRIMP[a] vs. maximal aerobic speed	$r = 0.66$, $P \leq 0.05$*
	Banister's TRIMP[a] vs. running economy	$r = 0.60$, $P \leq 0.05$*
	Banister's TRIMP[a] vs. Yo-Yo IR1 performance	$r = 0.65$, $P \leq 0.05$*
	Banister's TRIMP[a] vs. Yo-Yo IR2 performance	$r = 0.46$, $P > 0.05$
	Banister's TRIMP[a] vs. repeat-sprint test time	$r = 0.24$, $P > 0.05$
	Banister's TRIMP[a] vs. 5-m sprint time	$r = 0.28$, $P > 0.05$
	Banister's TRIMP[a] vs. 10-m sprint time	$r = 0.23$, $P > 0.05$
	Banister's TRIMP[a] vs. 20-m sprint time	$r = 0.35$, $P > 0.05$
	Edwards' TRIMP[a] vs. velocity at LT	$r = 0.19$, $P > 0.05$
	Edwards' TRIMP[a] vs. velocity at OBLA	$r = 0.38$, $P > 0.05$
	Edwards' TRIMP[a] vs. VO_{2max}	$r = 0.16$, $P > 0.05$
	Edwards' TRIMP[a] vs. velocity at VO_{2max}	$r = 0.38$, $P > 0.05$
	Edwards' TRIMP[a] vs. maximal aerobic speed	$r = 0.37$, $P > 0.05$

Table 8 (continued)

Study	Measures	Associations between training load and performance
	Edwards' TRIMP[a] vs. running economy	$r=0.34$, $P>0.05$
	Edwards' TRIMP[a] vs. Yo-Yo IR1 performance	$r=0.15$, $P>005$
	Edwards' TRIMP[a] vs. Yo-Yo IR2 performance	$r=0.21$, $P>0.05$
	Edwards' TRIMP[a] vs. repeat-sprint test time	$r=0.34$, $P>0.05$
	Edwards' TRIMP[a] vs. 5-m sprint time	$r=0.25$, $P>0.05$
	Edwards' TRIMP[a] vs. 10-m sprint time	$r=0.22$, $P>0.05$
	Edwards' TRIMP[a] vs. 20-m sprint time	$r=0.14$, $P>005$
	Lucia's TRIMP[a] vs. velocity at LT	$r=0.28$, $P>0.05$
	Lucia's TRIMP[a] vs. velocity at OBLA	$r=0.21$, $P>0.05$
	Lucia's TRIMP[a] vs. VO_{2max}	$r=0.61$, $P\leq0.05$*
	Lucia's TRIMP[a] vs. velocity at VO_{2max}	$r=0.41$, $P>0.05$
	Lucia's TRIMP[a] vs. maximal aerobic speed	$r=0.47$, $P>0.05$
	Lucia's TRIMP[a] vs. running economy	$r=0.37$, $P>0.05$
	Lucia's TRIMP[a] vs. Yo-Yo IR1 performance	$r=0.11$, $P>0.05$
	Lucia's TRIMP[a] vs. Yo-Yo IR2 performance	$r=0.21$, $P>0.05$
	Lucia's TRIMP[a] vs. repeat-sprint test time	$r=0.18$, $P>0.05$
	Lucia's TRIMP[a] vs. 5-m sprint time	$r=0.20$, $P>0.05$
	Lucia's TRIMP[a] vs. 10-m sprint time	$r=0.21$, $P>0.05$
	Lucia's TRIMP[a] vs. 20-m sprint time	$r=0.33$, $P>0.05$
	Team TRIMP[a] vs. velocity at LT	$r=0.19$, $P>0.05$
	Team TRIMP[a] vs. velocity at OBLA	$r=0.28$, $P>005$
	Team TRIMP[a] vs. VO_{2max}	$r=0.22$, $P>0.05$
	Team TRIMP[a] vs. velocity at VO_{2max}	$r=0.40$, $P>0.05$
	Team TRIMP[a] vs. maximal aerobic speed	$r=0.46$, $P>0.05$
	Team TRIMP[a] vs. running economy	$r=0.33$, $P>0.05$
	Team TRIMP[a] vs. Yo-Yo IR1 performance	$r=0.21$, $P>0.05$
	Team TRIMP[a] vs. Yo-Yo IR2 performance	$r=0.33$, $P>0.05$
	Team TRIMP[a] vs. repeat-sprint test time	$r=0.32$, $P>0.05$
	Team TRIMP[a] vs. 5-m sprint time	$r=0.22$, $P>0.05$
	Team TRIMP[a] vs. 10-m sprint time	$r=0.12$, $P>0.05$
	Team TRIMP[a] vs. 20-m sprint time	$r=0.33$, $P>0.05$
	iTRIMP[a] vs. velocity at LT	$r=0.68$, $P\leq0.01$*
	iTRIMP[a] vs. velocity at OBLA	$r=0.78$, $P\leq0.01$*
	iTRIMP[a] vs. VO_{2max}	$r=0.77$, $P\leq0.01$*
	iTRIMP[a] vs. velocity at VO_{2max}	$r=0.66$, $P\leq0.05$*
	iTRIMP[a] vs. maximal aerobic speed	$r=0.56$, $P\leq0.05$*
	iTRIMP[a] vs. running economy	$r=0.77$, $P\leq0.01$*
	iTRIMP[a] vs. Yo-Yo IR1 performance	$r=0.69$, $P\leq0.01$*
	iTRIMP[a] vs. Yo-Yo IR2 performance	$r=0.59$, $P\leq0.05$*
	iTRIMP[a] vs. repeat-sprint test time	$r=0.66$, $P\leq0.01$*
	iTRIMP[a] vs. 5-m sprint time	$r=0.54$, $P>0.05$
	iTRIMP[a] vs. 10-m sprint time	$r=0.48$, $P>0.05$
	iTRIMP[a] vs. 20-m sprint time	$r=0.66$, $P\leq0.01$*
Manzi et al. [35]	iTRIMP[a] vs. VO_2 at VT	$r=0.78$, very large, $P=0.002$*
	iTRIMP[a] vs. speed at [BLa^{-1}] of 4 mmol·L^{-1}	$r=0.64$, large, $P=0.004$*
	iTRIMP[a] vs. VO_{2max}	$r=0.77$, very large, $P=0.002$*
	iTRIMP[a] vs. Yo-Yo IRT (level 1) performance	$r=0.69$, large, $P=0.009$*
McCaskie et al. [30]	*Entire preseason (all players):*	
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P>0.05$

Table 8 (continued)

Study	Measures	Associations between training load and performance
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = 0.37$, moderate, $P = 0.02$*
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$r = -0.33$, moderate, $P = 0.03$*
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = -0.33$, moderate, $P = 0.03$*
	Last 11 weeks of the preseason (all players):	
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = 0.536$, large, $P = 0.001$*
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = -0.37$, moderate, $P = 0.02$*
	Entire preseason (midfielders/nomadic players only):	
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = 0.49$, moderate, $P = 0.03$*
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = -0.315$, moderate, $P = 0.18$
	Last 11 weeks of the preseason (midfielders/nomadic players only):	
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total on-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = 0.50$, moderate, $P = 0.03$*
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (entire season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first half of season)	$P > 0.05$
	Total off-field sRPE[a] vs. Champion Data© player rank[a] (first four games)	$r = -0.34$, moderate, $P = 0.15$
Nakamura et al. [38]	Total sRPE[a] vs. squat jump height	$r = -0.15$, small, $P > 0.05$
	Total sRPE[a] vs. CMJ height	$r = -0.16$, small, $P > 0.05$
	Total sRPE[a] vs. mean propulsive power relative to body mass	$r = 0.43$, moderate, $P > 0.05$
	Total sRPE[a] vs. 5-m velocity	$r = -0.18$, small, $P > 0.05$
	Total sRPE[a] vs. 10-m velocity	$r = -0.26$, small, $P > 0.05$
	Total sRPE[a] vs. 20-m velocity	$r = -0.27$, small, $P > 0.05$
	2-week sRPE[a] vs. squat jump height	$r = 0.27$, small, $P > 0.05$
	2-week sRPE[a] vs. CMJ height	$r = 0.25$, small, $P > 0.05$
	2-week sRPE[a] vs. mean propulsive power relative to body mass	$r = 0.09$, trivial, $P > 0.05$
	2-week sRPE[a] vs. 5-m velocity	$r = 0.48$, moderate, $P > 0.05$
	2-week sRPE[a] vs. 10-m velocity	$r = 0.60$, large, $P = 0.07$
	2-week sRPE[a] vs. 20-m velocity	$r = 0.66$, large, $P \leq 0.05$*
Stagno et al. [11]	Team TRIMP[a] vs. velocity at OBLA	$r = 0.71$, $P = 0.024$*
	Team TRIMP[a] vs. peak running speed	$P > 0.05$
	Team TRIMP[a] vs. time to exhaustion	$P > 0.05$
	Team TRIMP[a] vs. VO_{2max}	$r = 0.80$, $P = 0.017$*
	Mean weekly high-intensity activity vs. velocity at OBLA	$r = 0.67$, $P = 0.034$*
	Mean weekly high-intensity activity vs. peak running speed	$P > 0.05$
	Mean weekly high-intensity activity vs. time to exhaustion	$P > 0.05$
	Mean weekly high-intensity activity vs. VO_{2max}	$r = 0.65$, $P = 0.041$*
Taylor et al. [9]	sRPE[a] vs. velocity at LT	$R^2 = 0.11$, small, $P = 0.66$
	sRPE[a] vs. velocity at OBLA	$R^2 = 0.07$, Trivial, $P = 0.77$
	sRPE[a] vs. velocity at VO_{2max}	$R^2 = 0.14$, small, $P = 0.59$

Table 8 (continued)

Study	Measures	Associations between training load and performance
	sRPE[a] vs. VO_{2max}	$R^2=0.12$, small, $P=0.65$
	iTRIMP[a] vs. velocity at LT	$R^2=0.22$, small, $P=0.41$
	iTRIMP[a] vs. velocity at OBLA	$R^2=0.04$, Trivial, $P=0.93$
	iTRIMP[a] vs. velocity at VO_{2max}	$R^2=0.15$, small, $P=0.56$
	iTRIMP[a] vs. VO_{2max}	$R^2=0.55$, large, $P=0.06$
	Lucia's TRIMP[a] vs. velocity at LT	$R^2=0.20$, small, $P=0.46$
	Lucia's TRIMP[a] vs. velocity at OBLA	$R^2=0.02$, Trivial, $P=0.93$
	Lucia's TRIMP[a] vs. velocity at VO_{2max}	$R^2=0.49$, moderate, $P=0.10$
	Lucia's TRIMP[a] vs. VO_{2max}	$R^2=0.30$, moderate, $P=0.29$
	Edwards' TRIMP[a] vs. velocity at LT	$R^2=0.11$, small, $P=0.65$
	Edwards' TRIMP[a] vs. velocity at OBLA	$R^2=0.27$, small, $P=0.34$
	Edwards' TRIMP[a] vs. velocity at VO_{2max}	$R^2=0.02$, Trivial, $P=0.93$
	Edwards' TRIMP[a] vs. VO_{2max}	$R^2=0.40$, moderate, $P=0.17$
	Banister's TRIMP[a] vs. velocity at LT	$R^2=0.31$, moderate, $P=0.26$
	Banister's TRIMP[a] vs. velocity at OBLA	$R^2=0.21$, small, $P=0.43$
	Banister's TRIMP[a] vs. velocity at VO_{2max}	$R^2=0.26$, small, $P=0.34$
	Banister's TRIMP[a] vs. VO_{2max}	$R^2=0.78$, very large, $P=0.005$*

[BLa^{-1}] blood lactate concentration, *1RM* one-repetition maximum, *1RM$_{BP}$* one-repetition maximum bench press, *AD* load acceleration and deceleration distance *AU* arbitrary units, *CMJ* countermovement jump, *COD* change of direction, *HR* heart rate, *HR$_{max}$* maximum heart rate, *HSD* high-speed distance, *IFT* intermittent fitness test, *IRT* intermittent recovery test, *ISRT* intermittent shuttle run test, *iTRIMP* individualized training impulse, *LT* lactate threshold, *m > 30ASR* distance covered above 30% of anaerobic speed reserve, *m > MAS* distance covered above maximal aerobic speed, *OBLA* onset of blood lactate accumulation, *SD* standard deviation, *sRPE* session rating of perceived exertion, *t > 30ASR* time spent above 30% of anaerobic speed reserve, *t > MAS* time spent above maximal aerobic speed, *TRIMP* training impulse, *VHSD* very high-speed distance, *VO$_2$* oxygen uptake, *VO$_{2max}$* maximal oxygen uptake, *VT* ventilatory threshold, *W* watts, *W·kg^{-1}* watts per kilogram

[a]Formula is provided in Table3

3.6.3 Training Impulse Models and Aerobic Performance

Correlations between the individualized training impulse (iTRIMP) and changes in aerobic performance were assessed in four studies [3, 34–36]. In three studies [34–36], all correlations were significant and either ranged from large to very large or interpretations were not provided. In contrast, Akubat et al. [3] reported a significant, large correlation between iTRIMP and velocity at lactate threshold (LT) and nonsignificant small to moderate correlations between iTRIMP, velocity at LT, HR at LT, velocity at onset of blood lactate accumulation (OBLA), and HR at OBLA. One study [9] also reported nonsignificant, trivial to large curvilinear relationships between iTRIMP and aerobic performance.

Three studies [3, 11, 36] assessed correlations between the team training impulse (team TRIMP) and changes in aerobic performance. Stagno et al. [11] reported significant correlations between team TRIMP, velocity at OBLA, and maximal oxygen uptake (VO_{2max}) and nonsignificant correlations between team TRIMP, time to exhaustion, and peak running speed during the maximal treadmill test; no interpretations were provided. In contrast, Akubat et al. [3] and Malone et al. [36] reported nonsignificant, small to moderate correlations between team TRIMP and changes in aerobic performance.

The relationship between Banister's TRIMP and changes in aerobic performance was investigated in three studies [3, 9, 36]. Taylor et al. [9] reported a significant, very large curvilinear (quadratic) relationship between Banister's TRIMP and VO_{2max}, whereas curvilinear relationships with other aerobic variables (velocity at LT, velocity at OBLA, and velocity at VO_{2max}) were nonsignificant and small to moderate. Malone et al. [36] reported significant correlations between Banister's TRIMP and maximal aerobic speed, running economy, and Yo-Yo Intermittent Recovery Test (IRT) level 1 (IR1) performance, whereas correlations with other aerobic variables (i.e., velocity at LT, velocity at OBLA, VO_{2max}, velocity at VO_{2max}, Yo-Yo IR2 performance, and repeat-sprint ability) were nonsignificant, and no interpretations were provided. In contrast, Akubat et al. [3] reported nonsignificant, small to moderate correlations between Banister's TRIMP and changes in aerobic performance (i.e., velocity at LT, HR at LT, velocity at OBLA, and HR at OBLA).

The relationships between Edwards' TRIMP and changes in aerobic performance were investigated in four studies [7, 9, 27, 36]. Nonsignificant (or not reported) unclear to

large correlations between Edwards' TRIMP (total, during training, and during practice games) and changes in aerobic performance were evident in three studies [7, 27, 36]. In contrast, Taylor et al. [9] reported nonsignificant, trivial to moderate curvilinear relationships between Edwards' TRIMP and changes in aerobic performance.

Two studies [9, 36] investigated the relationship between Lucia's TRIMP and aerobic performance. Taylor et al. [9] reported nonsignificant, trivial to moderate curvilinear relationships between Lucia's TRIMP and changes in aerobic performance. Malone et al. [36] reported a significant correlation between Lucia's TRIMP and VO_{2max}, whereas correlations with other aerobic variables (velocity at LT, velocity at OBLA, velocity at VO_{2max}, maximal aerobic speed, running economy, Yo-Yo IR1 and IR2 performance, and repeat-sprint ability) were nonsignificant, and no interpretations were provided.

3.6.4 Training Impulse Models and Neuromuscular Variables

Relationships between training impulse models and changes in neuromuscular variables were only investigated in one study [36]. Malone et al. [36] reported nonsignificant correlations between Banister's TRIMP, Edwards' TRIMP, Lucia's TRIMP, team TRIMP and changes in neuromuscular variables, and no interpretations were provided. The correlation between iTRIMP and 20-m sprint time was significant, whereas correlations with other neuromuscular variables (i.e., 5-m and 10-m sprint time) were nonsignificant, and no interpretations were provided [35].

3.6.5 Perceptual Training Load Measures and Performance

In eight studies [3, 7, 9, 27, 36–40], correlations between total session rating of perceived exertion (sRPE) and changes in aerobic performance were assessed. Nonsignificant (or significance was not reported), trivial to very large correlations were identified in seven studies [3, 7, 36–39]. In contrast, Harrison and Johnston [39] reported a significant correlation between total sRPE and aerobic performance (2-km time trial).

The relationships between total sRPE and changes in neuromuscular variables were assessed in five studies [36–39, 41]. Gabbett and Domrow [37] reported a significant relationship between total sRPE and agility ("L" test) test time during the early-competition phase, whereas all other relationships between sRPE and neuromuscular variables (i.e., 10-, 20-, and 40-m sprint speed) during the preseason, early-competition, and late-competition phases were nonsignificant, and interpretations were not reported [37]. Similarly, nonsignificant (or not reported), unclear to large correlations between total sRPE and changes in neuromuscular variables were evident in four studies [36, 38, 39, 41]. Nakamura et al. [38] also reported correlations between sRPE for the first 2 weeks of the preseason and changes in neuromuscular variables. The correlation between 2-week sRPE and 20-m sprint velocity was significant and large, whereas the correlations between 2-week sRPE and all other neuromuscular variables (i.e., squat jump height, CMJ height, mean propulsive power relative to body mass, and 5- and 10-m sprint velocity) were nonsignificant and trivial to large [38]. Finally, Dobbin et al. [39] reported sRPE separately for resistance training, conditioning training, and skills training to determine correlations and R^2 with changes in neuromuscular variables. Correlations ranged from unclear to large, R^2 ranged from 0.00 to 0.62, and statistical significance was not reported [39].

The association between sRPE and game-related statistics was assessed in two studies [30, 41]. McCaskie et al. [30] investigated whether sRPE (on-field, off-field, and total) for all players or midfielders/nomadic players during the entire preseason or last 11 weeks of the preseason were associated with the Champion Data© player rank (first four games, first half of the season, and entire season). For all players, the correlations between on-field sRPE during the entire preseason, off-field sRPE for the entire preseason, on-field sRPE in the last 11 weeks of the preseason, off-field sRPE during the last 11 weeks of the preseason, and Champion Data© player rank for the first four games were significant and moderate to large [30]. For midfielders/nomadic players, correlations between on-field sRPE during the entire preseason, on-field sRPE during the last 11 weeks of the preseason, and Champion Data© player rank were significant and moderate [30]. The correlation between off-field sRPE for the entire preseason and Champion Data© player rank for the first half of the season was significant and moderate. All other correlations were nonsignificant and trivial to moderate [30]. Aughey et al. [42] used RPE to determine weekly and monthly training load (sRPE), training–stress balance, and training strain (see formulae in Table 3) and used effect sizes to compare these outcomes between wins and losses, with and without the covariates of days between games and ladder position. Statistical significance was not reported, and effect sizes ranged from small to moderate [42].

In four studies [23–26], RPE was reported separately for respiratory and leg musculature effort to determine respiratory and muscular sRPE. All studies [23–26] examined correlations between respiratory and muscular sRPE and changes in aerobic and neuromuscular variables. Correlations between respiratory sRPE and aerobic performance were nonsignificant (or not reported) and unclear to very large in three studies [23, 25, 26], and the results of the correlations were not reported in one study [24]. Correlations between respiratory sRPE and neuromuscular variables were nonsignificant (or not reported) and unclear to

large in three studies [23, 25, 26], and one study [24] did not provide correlation magnitudes and interpretation. A significant large correlation between muscular sRPE and aerobic performance ([BLa^{-1}] at 13 $km \cdot h^{-1}$) was evident in one study [25], and nonsignificant (or not reported) unclear to large correlations between muscular sRPE and aerobic performance (i.e., velocity at [BLa^{-1}] of 3 $mmol \cdot L^{-1}$, [BLa^{-1}] at 12 $km \cdot h^{-1}$, [BLa^{-1}] at 13 $km \cdot h^{-1}$, and time to exhaustion) were reported in three studies [23, 25, 26]. One study [24] did not report the correlations between muscular sRPE and aerobic performance. The correlations between muscular sRPE and neuromuscular performance (CMJ height on dominant and nondominant limbs) were significant and large in one study [25], and another study [24] reported a significant, large correlation between muscular sRPE and 15-m sprint time. Correlations between muscular sRPE and other neuromuscular variables (i.e., CMJ height, CMJ with arm swing, 5- and 15-m sprint times) were nonsignificant (or significance was not reported) and unclear to moderate in three studies [23, 24, 26].

4 Discussion

The importance of prescribing appropriate training loads to improve player performance is well recognized in the current literature [1, 10, 43]. However, there remains a lack of clarity regarding the most effective method for monitoring training load and the training load values most likely to promote improvements in player performance. Specifically, training loads that are too low will not facilitate positive training adaptations or can induce a detraining effect [1, 10]. In contrast, excessive loading can cause maladaptive responses to training, eventually resulting in diminished performance and other negative outcomes for players, such as illness or injury [16]. Therefore, the purpose of this review was to describe the associations between training load and performance outcomes in team sports. To effectively capture relevant literature, performance was considered to be objective indicators of in-game performance (e.g., game-related statistics) or changes in physical or physiological assessments across the monitoring period (e.g., aerobic fitness). As such, this review and its findings are important to ensure team sport coaching staff and practitioners make informed decisions on how to monitor and prescribe player training loads by establishing a fundamental understanding of the association between training load and performance.

4.1 External Training Load and Performance

Of the 12 articles that reported the associations between external training load and performance, seven reported external training load as exposure only. There were no clear associations between total exposure and performance (game-related statistics, aerobic performance, or neuromuscular variables), with significant and nonsignificant findings reported spanning a range of magnitudes. Reporting exposure for different seasonal phases or training modes did not make exposure more sensitive to changes in performance.

A likely explanation for the inconsistent associations between exposure and performance (game-related statistics, aerobic performance, and neuromuscular variables) outcomes within and between studies is the limited detail that exposure provides. In essence, exposure is a measure of duration and does not objectively quantify the physical demands or intensity of training sessions or games. Using only exposure to quantify external training load also limits the ability to detect individual differences in external training load between players. Specifically, if all players complete the same training session or play the entire game, their external training load value will be identical despite working at different intensities. Therefore, when the limitations of exposure are considered, the findings of this review suggest that more detailed, individualized measures of external training load should be implemented when monitoring external training load to optimize team sport performance.

Only three [9, 27, 30] of the studies included in this review examined alternative external training load measures beyond exposure. Fitzpatrick et al. [27] reported moderate to very large correlations between a range of GPS variables and changes in aerobic performance; while five of these correlations were moderate to very large, the remainder (15 correlations) were unclear (see Table 7). Similarly, Taylor et al. [9] reported nonsignificant correlations between total distance, individualized high-speed distance, high-speed distance, very high-speed distance, PlayerLoad™, and changes in aerobic performance. However, as only ten players were monitored, it is possible that the sample size was insufficient to accurately detect statistical significance. As such, the magnitude and qualitative interpretation of the relationships should be considered as equally important in the context of the investigation by Taylor et al. [9]. Only total distance and very high-speed distance showed large correlations with performance. These findings are consistent with previous literature that advocates for the concurrent monitoring of training volume and intensity as they provide different information regarding the demands of training and competition [2, 44]. An important finding reported by Taylor et al. [9] is that PlayerLoad™ possessed only a trivial association with aerobic performance, despite being considered an alternative measure of training volume. As such, it is important for coaches and practitioners to recognize that, when utilizing commercially available microsensor technologies, GPS and accelerometer-derived measures of training volume may not be interchangeable.

Only McCaskie et al. [30] investigated the association between external load measures beyond exposure and game-related statistics. They reported significant, moderate correlations between selected GPS variables, PlayerLoad™ and Champion Data© player rank during the first four games of the season. Furthermore, only nonsignificant, trivial to small correlations between external training load during the preseason and performance during the first half or entire season were reported. The inclusion of objective variables rather than exposure is preferable; however, McCaskie et al. [30] collected training load data during the preseason only. Collecting data in the preseason only does not account for the training demands experienced throughout the entire season and the extent that this likely influences game-related statistics.

4.2 Internal Training Load and Performance

Internal training load represents the physical and physiological adaptations of players in response to the imposed demands. Therefore, it is the internal training load that will ultimately dictate training outcomes [2, 4, 6]. As such, external load should be manipulated to encourage players to reach the internal load values needed to promote favorable performance outcomes. The range of outcome measures reported for both internal load and performance make between-study comparisons difficult; nonetheless, some training load measures, such as the volume of high-intensity activity performed, consistently showed significant and large to very large associations with aerobic performance.

4.2.1 Training Intensity and Performance

The findings of this review show significant and large associations between the volume of high-intensity activity completed, defined as time spent above 90% of maximum HR (HR_{max}), and changes in aerobic performance. This finding is in agreement with previous literature documenting the greater effectiveness of increasing the intensity of the activity rather than volume to promote improvements in aerobic capacity [45]. Working at intensities at or close to VO_{2max} is necessary to maximize oxygen transport and utilization systems to attain near maximal cardiac output [45]. By increasing the efficiency of oxygen delivery, players can subsequently sustain longer bouts of high-intensity activity and recover more quickly between these bouts [46]. The large associations between high-intensity activity and aerobic performance also support the need for an individualized rather than a fixed approach to monitor internal training load intensity in team sports, as all studies reporting intensity distribution used individualized cutpoints based on each player's maximum HR. However, it should be noted that, despite the consistently strong associations between high-intensity activity and aerobic performance, all of the relevant studies in this review were conducted in field-based sports. Previous work has noted that, in court-based sports, which generally involve a higher proportion of lateral activity and more frequent change of direction, independent HR measurement is likely to underestimate internal training load as a result of rapid changes in intensity [47–49]. Therefore, more work is needed to determine whether the results of this review can be reasonably applied to court-based team sports.

4.2.2 Heart Rate-Mediated Training Load Models and Performance

HR-mediated training load models are common in team sports; however, the varied relationships between each training load model and aerobic performance identified in this review suggest the various models may not be interchangeable. Banister's, Lucia's, and iTRIMP models are all predicated on the relationship between HR and [BLa^{-1}] during incremental exercise. Despite being conceptually similar, only the iTRIMP consistently showed significant and large to very large associations with changes in aerobic performance. A possible explanation for the inconsistent associations between HR-mediated training load models and aerobic performance is that these models generally assume a fixed response to exercise obtained during a maximal, graded exercise test [2]. Therefore, these models may not accurately reflect the intermittent nature of team sports, which can comprise numerous bouts of high-intensity activity and other movements such as changes in direction, accelerations, decelerations, and jumps. For this reason, it is important to determine the suitability of various training load models to capture exercise volume and intensity as they apply to specific team sports. Alternatively, the stronger correlations between iTRIMP and changes in aerobic performance compared with Banister's and Lucia's TRIMP is likely because the iTRIMP is the only fully individualized training load model. Using iTRIMP, the HR and blood lactate responses to incremental exercise are determined for each player and then used to develop weighting factors. These individualized weighting factors are crucial given the large variation in the HR–blood lactate relationship between individuals [49]. Therefore, this finding provides further support for the use of individualized measures of internal load; however, as with all training load models, it is important to consider the suitability of the approach for measuring specific outcomes as well as how easily the approach can be implemented in practice. In this regard, the iTRIMP has several practical limitations that practitioners should recognize. First, to determine the relationship between HR and blood lactate, each player must complete a maximal, graded exercise test; specialized expertise can be required to run these tests, and they can be time and resource intensive. In addition, the testing needs

to be repeated regularly as changes in the HR–blood lactate relationship will vary with changes in anaerobic and aerobic performance across various stages of the annual plan [2].

In this review, HR-mediated high-intensity activity consistently showed significant and large or very large associations with the same aerobic variables as iTRIMP (e.g., speeds at [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$, VO_{2max}, and Yo-Yo IRT performance). Therefore, these findings suggest that iTRIMP may be no more sensitive at detecting changes in aerobic performance than HR-mediated high-intensity activity. Given the practical limitations of iTRIMP, use of HR-mediated intensity zones rather than the iTRIMP may be more suitable to anticipate the aerobic performance of players in field-based team sports.

4.2.3 Perceptual Training Load and Performance

Despite the presence of some significant and large to very large correlations between sRPE and changes in aerobic and neuromuscular variables, the studies included in this review predominantly reported nonsignificant and trivial to moderate correlations between sRPE and game-related statistics, as well as changes in aerobic and neuromuscular variables. Of the studies that reported associations between sRPE and performance ($N = 16$), only six reported significant associations [24, 25, 30, 37, 38, 40], and only eight [7, 23–26, 30, 38, 39] reported large or very large associations. Further, these associations were only between selected variables or time points and not consistent within or between studies. As a whole, these findings suggest perceptual training load measures have limited association with player performance.

It has been previously highlighted that sRPE is influenced by a range of factors such as stress during competition and the type of training completed (e.g., conditioning vs. games-based training) [44, 49]. To negate these external influences, methods to make sRPE more accurate have been proposed, such as reporting aerobic and neuromuscular perceived exertion separately [24–26]. The findings of this review suggest the use of differential sRPE (respiratory and neuromuscular) is more sensitive to changes in aerobic and neuromuscular variables than total sRPE. However, the inconsistencies in the results (i.e., significant and nonsignificant correlations spanning a range of magnitudes) and how the data have been analyzed (e.g., differences in statistical tests utilized) means that there is limited evidence to support the accuracy of sRPE to predict performance outcomes. This review focusses on the associations between training load measures and performance, and these findings may not hold true for other applications of monitoring (e.g., fatigue, injury risk, or illness). The different mechanisms underpinning various player responses makes it possible that a single training load measure may not be sufficient to capture all of the variables of interest to coaches or practitioners. Therefore, it is particularly important in practice to consider the purpose of training load monitoring when deciding on the best approach or combination of approaches to implement.

4.3 Limitations and Future Directions

The results of this review provide important considerations for training load and performance monitoring in team sports. However, there are limitations that must be considered when applying the findings. Of the articles that reported performance as changes in physical or physiological assessments, the monitoring period ranged from 4 weeks to two consecutive seasons. As such, it is possible that this variability contributed to the inconsistencies in the associations between training load and performance variables across studies. The length of the monitoring period will affect the aerobic and neuromuscular adaptations that take place in team sport players, influencing the performance outcomes observed. In regards to performance, the lack of in-game data must also be considered in the context of this review. Specifically, players were monitored during the in-season phase of the annual plan in a number of studies; however, external and internal training loads encountered during competition were not included in the analyses. This is a concern when analyzing and interpreting findings across studies, as total load accumulated across the week or across the monitoring period may be underreported. To negate this limitation, it would be preferable for future studies to monitor players during competition and include these data in the analyses; however, the feasibility of this practice must be considered. Often in-game monitoring is not possible because of the need to transport equipment where games are held at away venues or because of restrictions around the use of wearable technologies (e.g., HR monitors or GPS units) during competition. To counter this limitation, team sport researchers and practitioners might consider substituting normative training load values experienced during competition, such as those derived from practice or small-sided games. While training loads experienced during competition will vary, the inclusion of normative values are likely to make values reported closer to the actual weekly loads achieved than if competition data were omitted entirely.

The lack of in-game data is not only important for the studies in this review but should also be considered for future research. Success during competition is largely determined by a combination of components relative to each sport, such as anaerobic and aerobic fitness, explosive power, and repeat-sprint ability; however, most studies have assessed these factors separately. Therefore, the associations between training load and performance measures such as in-game workloads or game-related statistics should be more closely examined, as they may possess higher ecological validity compared with assessing individual performance

components separately. In addition, factors that influence in-game performance, such as decision making, team interaction, and mental fatigue [50], may not be adequately captured when physical or physiological tests are used as the sole indicators of player performance. The associations between training load and the demands of competitive games is currently lacking from the literature, but this information is fundamental to understanding the extent to which pre- and in-season training loads are effective at preparing players for competition. While most coaches and practitioners assume that the training loads delivered leading into competition are sufficient [44], very rarely is this assumption confirmed. Consequently, it would be of great value to researchers, coaches, and practitioners to determine the extent that training loads relate to in-game workloads. Alternatively, assessing game-related statistics is advantageous from a practical perspective given they can be collected with no additional burden on players and are generally freely accessible. Only three of the articles in this review reported game-related statistics [8, 42], with one reporting training load as exposure only [8] and the other sRPE only [23]. Given that exposure and sRPE showed consistently weak associations with other performance variables across this review, it is possible that these studies [8, 42] do not well-represent the associations between training load and in-game performance. While one study [30] reported accelerometer- and GPS-derived external load, training load data were only collected during the preseason and do not account for in-season training demands and how this affects in-game performance. Alternatively, the weak relationships between training load and game-related statistics could be partially attributed to the complexity of team sport performance. For example, game-related contextual variables such as ladder position, level of opposition, and tactical strategies employed will influence the external and internal training loads encountered during games as well as the game outcome (e.g., win or loss) [42]. Therefore, more work should be conducted to understand how common training load models relate to performance, as measured by in-game data or game-related statistics and the extent that this relationship is influenced by game-related contextual variables.

In examining the included studies in this review, it was apparent the type of statistical analyses conducted varied widely, making it difficult to conduct direct comparisons between studies. First, several studies only conducted analyses to determine either the practical (e.g., effect sizes) or the statistical (e.g., correlations) significance of the data in isolation. Given the small sample sizes that are common in team sport studies, including in many of the studies in this review, the practical significance and correlation magnitudes of the data should be considered equally important to the statistical significance. It would also be helpful in future studies if the results of sample size calculations were reported to create transparency around whether the sample was sufficient to detect statistical significance. Furthermore, to facilitate between-study comparisons, analyses to detect statistical and practical significance should be conducted (assuming the sample size is sufficient) and correlation magnitudes and interpretations should be provided. In addition, correlation magnitudes in this review were reported based on the cutpoints identified in each study (where applicable). This approach was adopted to avoid any bias in reporting on behalf of the research team; however, this should be recognized when comparing study findings.

Most studies ($N = 23$) examined linear relationships between training load and performance variables, possibly because of the simplicity of these analyses. In team sports, training intensity is predominantly captured at low and high intensities [31, 32], which—when combined with the findings of this review that high-intensity activity is most strongly correlated with changes in aerobic performance—suggests that the relationships between training load and aerobic capacity is likely to be curvilinear. There is a clear contrast between studies examining linear and curvilinear associations between training load and performance. For example, four studies [3, 34–36] reported significant, linear associations between iTRIMP and changes in aerobic performance, while Taylor et al. [9] reported nonsignificant curvilinear associations between iTRIMP and changes in aerobic performance. Given the lack of comparative data examining curvilinear relationships between training load and performance, it is difficult to preference one type of analysis over another; however, future research should consider comparing the nature of the relationship or using curve fittings to better understand the utility of common training load models to detect performance capacity.

Although correlation analyses are common in training load studies, the limited capacity for training load to predict injuries in team sports despite the presence of strong correlations was recently reported [51]. While injury and performance prediction may not be best detected by the same training load model, it is important to determine the presence of causality when attempting to monitor or predict player performance. The nature of team sport research means that prediction (e.g., regression) analyses are also likely to be more suited to the data being collected. Specifically, utilizing appropriate regression analyses will take into account multiple observations per participant, which are commonplace when data are being collected across multiple training sessions or games [10, 51, 52]. In studies where multiple training load measures are being reported, certain regression models (e.g., multilevel modelling) are also more suited to the data as they do not treat variables independent of one another and can account for associations between the independent variables (e.g., internal and external training load) [10, 52, 53]. One limitation with common regression models

is the need for larger sample sizes to be able to detect statistical significance because of the number of training load variables being assessed together. Therefore, analyses such as principal component analyses have the potential to be used to understand the relationships between training load variables and determine which outcome measures provide the same and unique information [10]. For example, Weaving et al. [53] recently utilized principal component analyses in rugby union to identify the combination of external and internal training load metrics (total distance, high-speed distance, PlayerLoad™, and sRPE) able to capture the most information (variance). This study determined that, for most players, PlayerLoad™, total distance, or sRPE, as well as high-speed distance, could be used for quantifying training load in rugby union during skills training, rather than all variables being concurrently monitored, as is often the case in team sports [53]. Therefore, principal component analyses should be explored further in team sports to limit the number of variables included in analyses such as regression models, thus lowering the necessary sample size and allowing for the utilization of statistical analyses more suited to the type of data collected [54].

5 Conclusions

This review was the first to provide a systematic evaluation of the association between training load and performance in team sports. Internal training load showed stronger associations with performance than external training load, but these associations varied based on the measures investigated. In all studies, HR-derived high-intensity activity was strongly associated with aerobic performance, predominantly speed at [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$, VO_{2max}, and intermittent endurance capacity (IFT and Yo-Yo IRT performance). sRPE and Banister's, Edwards', and Lucia's TRIMP models showed consistently weak associations with performance.

6 Practical Applications

The findings of this review pose significant practical applications that team sport practitioners can use to optimize training load monitoring to track the performance capacity of players. When seeking to predict player performance, practitioners should use internal training load measures. However, an understanding of the relationship between external and internal training loads is necessary to allow practitioners to effectively modify the training stimulus (external training load) if changes in internal training load are required [2, 4, 6]. Where possible, external load measures such as speed and distance should be monitored, as exposure does not provide adequate information regarding the demands or intensity of training and competition. When seeking to promote improvements in aerobic capacity (e.g., speeds at [BLa^{-1}] of 2 and 4 $mmol \cdot L^{-1}$, VO_{2max}, and Yo-Yo IRT performance), coaches and practitioners should aim to implement and monitor bouts of high-intensity activity ($\geq 90\%$ of HR_{max}). Practitioners should exercise caution when using common training load models (e.g., sRPE and Banister's, Lucia's, and Edwards' TRIMP) to detect performance in field-based sports.

Compliance with Ethical Standards

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